

Uniaxial Tension Testing

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THE TENSION TEST is one of the most commonly used tests for evaluating materials. In its simplest form, the tension test is accomplished by gripping opposite ends of a test item within the load frame of a test machine. A tensile force is applied by the machine, resulting in the gradual elongation and eventual fracture of the test item. During this process, force-extension data, a quantitative measure of how the test item deforms under the applied tensile force, usually are monitored and recorded. When properly conducted, the tension test provides force-extension data that can quantify several important mechanical properties of a material. These mechanical properties determined from tension tests include, but are not limited to, the following:

- Elastic deformation properties, such as the modulus of elasticity (Young's modulus) and Poisson's ratio
- Yield strength and ultimate tensile strength
- Ductility properties, such as elongation and reduction in area
- Strain-hardening characteristics

These material characteristics from tension tests are used for quality control in production, for ranking performance of structural materials, for evaluation of newly developed alloys, and for dealing with the static-strength requirements of design.

The basic principle of the tension test is quite simple, but numerous variables affect results. General sources of variation in mechanical-test results include several factors involving materials, namely, methodology, human factors, equipment, and ambient conditions, as shown in the "fish-bone" diagram in Fig. 1. This article discusses the methodology of the tension test and the effect of some of the variables on the tensile properties determined. The following methodology and variables are discussed:

- Shape of the item being tested
- Method of gripping the item
- Method of applying the force
- Determination of strength properties other than the maximum force required to fracture the test item
- Ductility properties to be determined

- Speed of force application or speed of elongation (e.g., control of stress rate or strain rate)
- Test temperature

The main focus of this article is on the methodology of tension tests as it applies to metallic materials. Factors associated with test machines and their method of force application are described in more detail in the article "Testing Machines and Strain Sensors" in this Volume.

This article does not address the tension testing of nonmetallic materials, such as plastics, elastomers, or ceramics. Although uniaxial tension testing is used in the mechanical evaluation of these materials, other test methods often are used for mechanical-property evaluation. The general concept of tensile properties is very similar for these nonmetallic materials, but there are also some very important differences in their behavior and the required test procedures for these materials:

- Tension-test results for plastics depend more strongly on the strain rate because plastics are viscoelastic materials that exhibit time-dependent deformation (i.e., creep) during force application. Plastics are also more sensitive to temperature than metals. Thus, control of strain rates and temperature are more critical with plastics, and sometimes tension tests are run at more than one strain and/or temperature. The ASTM standard for tension testing of plastics is D 638.
- Tension testing of ceramics requires more attention to alignment and gripping of the test piece* in the test machine because ceramics are brittle materials that are extremely sensitive to bending strains and because the hard surface of ceramics reduces the effectiveness of frictional gripping devices. The need for a large gripping areas thus requires the use of larger test pieces (Ref 1). The ASTM standard for tension testing of monolithic ceramics at room temperature is C 1275. The standard for continuous fiber-reinforced advanced ceramics at ambient temperatures is C 1273.

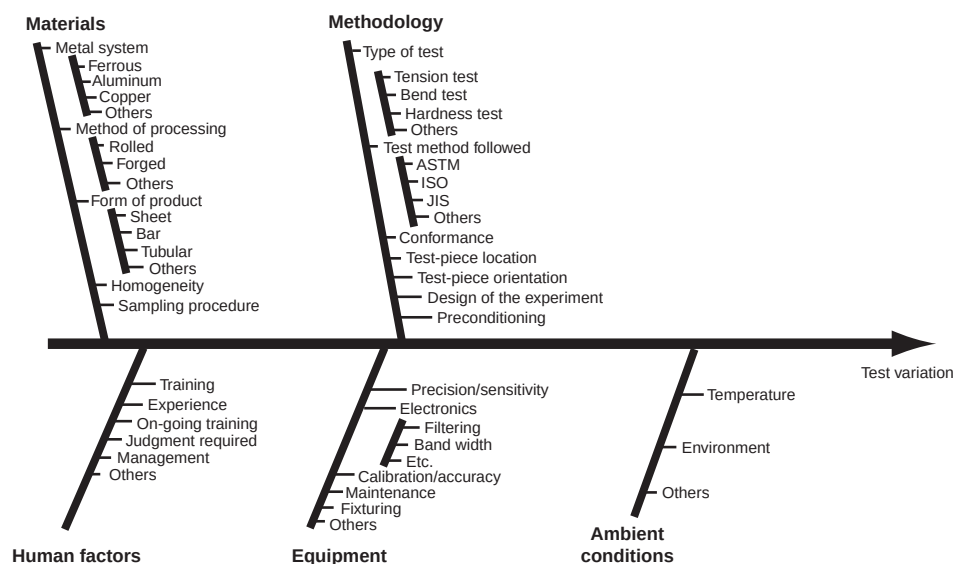


Fig. 1 "Fish-bone" diagram of sources of variability in mechanical-test results

*The term "test piece" is used in this article for what is often called a "specimen" (see "The Test Piece" in this article).

- *Tension testing of elastomers* is described in ASTM D 412 with specific instructions about test-piece preparation, equipment, and test conditions. Tensile properties of elastomers vary widely, depending on the particular formulation, and scatter both within and between laboratories is appreciable compared with the scatter of tensile-test results of metals (Ref 2). The use of tensile-test results of elastomers is limited principally to comparison of compound formulations.

Definitions and Terminology

The basic results of a tension test and other mechanical tests are quantities of stress and strain that are measured. These basic terms and their units are briefly defined here, along with discussions of basic stress-strain behavior and the differences between related terms, such as stress and force and strain and elongation.

Load (or force) typically refers to the force acting on a body. However, there is currently an effort within the technical community to replace the word load with the more precise term *force*, which has a distinct meaning for any type of force applied to a body. **Load applies, in a strict sense, only to the gravitational force that acts on a mass. Nonetheless, the two terms are often used interchangeably.**

Force is usually expressed in units of pounds-force, lbf, in the English system. In the metric system, force is expressed in units of newtons (N), where one newton is the force required to give a 1 kg mass an acceleration of 1 m/s² (1 N = 1 kgm/s²). Although newtons are the preferred metric unit, force is also expressed as kilogram force, kgf, which is the gravitational force on a 1 kg mass on the surface of the earth. The numerical conversions between the various units of force are as follows:

- 1 lbf = 4.448222 N or 1 N = 0.2248089 lbf
- 1 kgf = 9.80665 N

In some engineering disciplines, such as civil engineering, the quantity of 1000 lbf is also expressed in units of kip, such that 1 kip = 1000 lbf.

Stress is simply the amount of force that acts over a given cross-sectional area. Thus, stress is expressed in units of force per area units and is obtained by dividing the applied force by the cross-sectional area over which it acts. Stress is an important quantity because it allows strength comparison between tests conducted using test pieces of different sizes and/or shapes. When discussing strength values in terms of force, the load (force) carrying capacity of a test piece is a function of the size of the test piece. However, when material strength is defined in terms of stress, the size or shape of the test piece has little or no influence on stress measurements of strength (provided the cross section contains at least 10 to 15 metallurgical grains).

Stress is typically denoted by either the Greek symbol sigma, σ , or by s , unless a distinction is being made between true stress and nominal (engineering) stress as discussed in this article. The units of stress are typically lbf/in.² (psi) or thousands of psi (ksi) in the English system and a pascal (Pa) in the metric system. Engineering stresses in metric units are also expressed in terms of newtons per area (i.e., N/m² or N/mm²) or as kilopascals (kPa) and megapascals (MPa). Conversions between these various units of stress are as follows:

- 1 Pa = 1.45×10^{-4} psi
- 1 Pa = 1 N/m²
- 1 kPa = 10^3 Pa or 1 kPa = 0.145 psi
- 1 MPa = 10^6 Pa or 1 MPa = 0.145 ksi
- 1 N/mm² = 1 MPa

Strain and elongation are similar terms that define the amount of deformation from a given amount of applied stress. In general terms, strain is defined (by ASTM E 28) as “the change per unit length due to force in an original linear dimension.” The phrase *change per unit length* means that a change in length, ΔL , is expressed as a ratio of the original length, L_0 . This change in length can be expressed in general terms as a strain or as elongation of gage length, as described subsequently in the context of a tension test.

Strain is a general term that can be expressed mathematically, either as engineering strain or as true strain. Nominal (or engineering) strain is often represented by the letter e , and logarithmic (or true) strain is often represented by the Greek letter ϵ . The equation for engineering strain, e , is based on the nominal change in length, (ΔL) where:

$$e = \Delta L/L_0 = (L - L_0)/L_0$$

The equation for true strain, ϵ , is based on the instantaneous change in length (dl) where:

$$\epsilon = \int_{L_0}^L \frac{dl}{l} = \ln \left(\frac{L}{L_0} \right)$$

These two basic expressions for strain are interrelated, such that:

$$\epsilon = \ln(1 + e)$$

In a tension test, the typical measure of strain is engineering strain, e , and the units are inches per inch (or millimeter per millimeter and so on). Often, however, no units are shown because strain is the ratio of length in a given measuring system.

This article refers to only engineering strain unless otherwise specified. In a tension test, true strain is based on the change in the cross-sectional area of the test piece as it is loaded. It is not further discussed herein, but a detailed discussion is found in the article “Mechanical Behavior under Tensile and Compressive Loads” in this Volume.

Elongation is a term that describes the amount that the test-piece stretches during a

tension test. This stretching or elongation can be defined either as the total amount of stretch, ΔL , that a part undergoes or the increase in gage length per the initial gage length, L_0 .

The latter definition is synonymous with the meaning of engineering strain, $\Delta L/L_0$, while the first definition is the total amount of extension. Because two definitions are possible, it is imperative that the exact meaning of elongation be understood each time it is used.

This article uses the term *elongation, e* , to mean nominal or engineering strain (i.e., $e = \Delta L/L_0$). The amount of stretch is expressed as extension, or the symbol ΔL . In many cases, elongation, e , is also reported as a percentage change in gage length as a measure of ductility (i.e., percent elongation), $(\Delta L/L_0) \times 100$.

Engineering Stress and True Stress. Along with the previous descriptions of engineering strain and true strain, it is also possible to define stress in two different ways as engineering stress and true stress. As is intuitive, when a tensile force stretches a test piece, the cross-sectional area must decrease (because the overall volume of the test piece remains essentially constant). Hence, because the cross section of the test piece becomes smaller during a test, the value of stress depends on whether it is calculated based on the area of the unloaded test piece (the initial area) or on the area resulting from that applied force (the instantaneous area). Thus, in this context, there are two ways to define stress:

- **Engineering stress, s :** The force at any time during the test divided by the initial area of the test piece; $s = F/A_0$ where F is the force, and A_0 is the initial cross section of a test piece.
- **True stress, σ :** The force at any time divided by the instantaneous area of the test piece; $\sigma = F/A_i$ where F is the force, and A_i is the instantaneous cross section of a test piece.

Because an increasing force stretches a test piece, thus decreasing its cross-sectional area, the value of true stress will always be greater than the nominal, or engineering, stress.

These two definitions of stress are further related to one another in terms of the strain that occurs when the deformation is assumed to occur at a constant volume (as it frequently is). As previously noted, strain can be expressed as either engineering strain (e) or true strain, where the two expressions of strain are related as $\epsilon = \ln(1 + e)$. When the test-piece volume is constant during deformation (i.e., $A_i L_i = A_0 L_0$), then the instantaneous cross section, A_i , is related to the initial cross section, A_0 , where

$$A = A_0 \exp \{-\epsilon\}$$

$$= A_0 / (1 + e)$$

If these expressions for instantaneous and initial cross sections are divided into the applied force to obtain values of true stress (at the

instantaneous cross section, A_i) and engineering stress (at the initial cross section, A_0), then:

$$\sigma = s \exp \{\epsilon\} = s(1 + e)$$

Typically, engineering stress is more commonly considered during uniaxial tension tests. All discussions in this article are based on nominal engineering stress and strain unless otherwise noted. More detailed discussions on true stress and true strain are in the article “Mechanical Behavior under Tensile and Compressive Loads” in this Volume.

Stress-Strain Behavior

During a tension test, the force applied to the test piece and the amount of elongation of the test piece are measured simultaneously. The applied force is measured by the test machine or by accessory force-measuring devices. The amount of stretching (or extension) can be measured with an extensometer. An extensometer is a device used to measure the amount of stretch that occurs in a test piece. Because the amount of elastic stretch is quite small at or around the onset of yielding (in the order of 0.5% or less for steels), some manner of magnifying the stretch is required. An extensometer may be a mechanical device, in which case the magnification occurs by mechanical means. An extensometer may also be an electrical device, in which case the magnification may occur by mechanical means, electrical means, or by a combination of both. Extensometers generally have fixed gage lengths. If an extensometer is used only to obtain a portion of the stress-strain curve sufficient to determine the yield properties, the gage length of the extensometer may be shorter than the gage length required for the elongation-at-fracture measurement. It may also be longer, but in general, the extensometer gage length should not exceed approximately 85 to 90% of the length of the reduced section or of the distance between the grips for test pieces without reduced sections. This ratio for some of the most common test configurations with a 2 in. gage length and $\frac{2}{4}$ in. reduced section is 0.875%.

The applied force, F , and the extension, ΔL , are measured and recorded simultaneously at regular intervals, and the data pairs can be converted into a stress-strain diagram as shown in Fig. 2. The conversion from force-extension data to stress-strain properties is shown schematically in Fig. 2(a). Engineering stress, s , is obtained by dividing the applied force by the original cross-sectional area, A_0 , of the test piece, and strain, e , is obtained by dividing the amount of extension, ΔL , by the original gage length, L . The basic result is a stress-strain curve (Fig. 2b) with regions of elastic deformation and permanent (plastic) deformation at stresses greater than those of the elastic limit (EL in Fig. 2b).

Typical stress-strain curves for three types of steels, aluminum alloys, and plastics are shown in Fig. 3 (Ref 3). Stress-strain curves for some structural steels are shown in Fig. 4(a) (Ref 4) for elastic conditions and for small amounts of plastic deformation. The general shape of the stress-strain curves can be described for deformation in this region. However, as plastic deformation occurs, it is more difficult to generalize about the shape of the stress-strain curve. Figure 4(b) shows the curves of Fig. 4(a) continued to fracture.

Elastic deformation occurs in the initial portion of a stress-strain curve, where the stress-strain relationship is initially linear. In this region, the stress is proportional to strain. Mechanical behavior in this region of stress-strain curve is defined by a basic physical property called the **modulus of elasticity** (often abbreviated as E). The modulus of elasticity is the slope of the stress-strain line in this linear region, and it is a basic physical property of all materials. It essentially represents the spring constant of a material.

The modulus of elasticity is also called Hooke's modulus or Young's modulus after the scientists who discovered and extensively studied the elastic behavior of materials. The behavior was first discovered in the late 1600s by the English scientist Robert Hooke. He observed that a given force would always cause a repeatable, elastic deformation in all materials. He further discovered that there was a force above which the deformation was no longer elastic; that is, the material would not return to its original length after release of the force. This limiting force is called the **elastic limit** (EL in Fig. 2b). Later, in the early 1800s, Thomas Young, an English physicist, further investigated and described this elastic phenomenon, and so his name is associated with it.

The **proportional limit (PL)** is a point in the elastic region where the linear relationship between stress and strain begins to break down. At some point in the stress-strain curve (PL in Fig. 2b), linearity ceases, and small increase in stress causes a proportionally larger increase in strain. This point is referred to as the proportional limit (PL) because up to this point, the stress and strain are proportional. If an applied force below the PL point is removed, the trace of the stress and strain points returns along the original line. If the force is reapplied, the trace of the stress and strain points increases along the original line. (When an exception to this linearity is observed, it usually is due to mechanical hysteresis in the extensometer, the force indicating system, the recording system, or a combination of all three.)

The **elastic limit (EL)** is a very important property when performing a tension test. If the applied stresses are below the elastic limit, then the test can be stopped, the test piece unloaded, and the test restarted without damaging the test piece or adversely affecting the test results. For example, if it is observed that the extensometer is not recording, the force-elongation curve shows an increasing force, but no elongation. If

the force has not exceeded the elastic limit, the test piece can be unloaded, adjustments made, and the test restarted without affecting the results of the test. However, if the test piece has been stressed above the EL, plastic deformation (set) will have occurred (Fig. 2b), and there will be a permanent change in the stress-strain behavior of the test piece in subsequent tension (or compression) tests.

The PL and the EL are considered identical in most practical instances. In theory, however, the EL is considered to be slightly higher than the PL, as illustrated in Fig. 2b. The measured values of EL or PL are highly dependent on the magnification and sensitivity of the extensometer used to measure the extension of the test piece. In addition, the measurement of PL and EL also highly depends on the care with which a test is performed.

Plastic Deformation (Set) from Stresses above the Elastic Limit. If a test piece is stressed (or loaded) and then unloaded, any

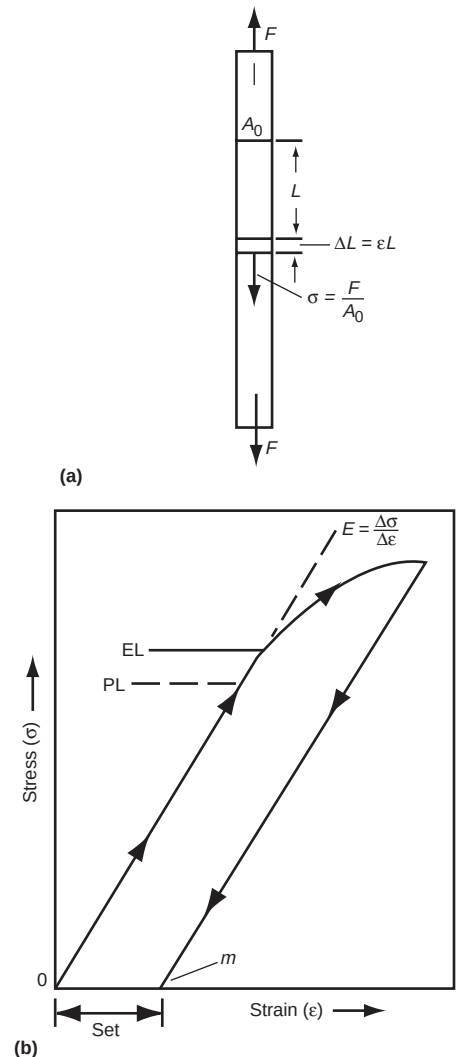


Fig. 2 Stress-strain behavior in the region of the elastic limit. (a) Definition of σ and ϵ in terms of initial test piece length, L , and cross-sectional area, A_0 , before application of a tensile force, F . (b) Stress-strain curve for small strains near the elastic limit (EL)

retest proceeds along the unloading path whether or not the elastic limit was exceeded. For example, if the initial stress is less than the elastic limit, the load-unload-reload paths are identical. However, if a test piece is stressed in tension beyond the elastic limit, then the unload path is offset and parallel to the original loading path (Fig. 2b). Moreover, any subsequent tension measurements will follow the previous unload path parallel to the original stress-strain line. Thus, the application and removal of stresses above the elastic limit affect all subsequent stress-strain measurements.

The term *set* refers to the permanent deformation that occurs when stresses exceed the elastic limit (Fig. 2b). ASTM E 6 defines *set* as the strain remaining after the complete release of a load-producing deformation. Because set is

permanent deformation, it affects subsequent stress-strain measurements whether the reloading occurs in tension or compression. Likewise, permanent set also affects all subsequent tests if the initial loading exceeds the elastic limit in compression. Discussions of these two situations follow.

Reloading after Exceeding the Elastic Limit in Tension. If a test piece is initially loaded in tension beyond the elastic limit and then unloaded, the unload path is parallel to the initial load path but offset by the set; on reloading in tension, the unloading path will be followed. Figure 5 illustrates a series of stress-strain curves obtained using a machined round test piece of steel. (The strain axis is not to scale.) In this figure, the test piece was loaded first to Point A and unloaded. The area of the test piece

was again determined (A_2) and reloaded to Point B and unloaded. The area of the test piece was determined for a third time (A_3) and reloaded until fracture occurred. Because during each loading the stresses at Points A and B were in excess of the elastic limit, plastic deformation occurred. As the test piece is elongated in this series of tests, the cross-sectional area must decrease because the volume of the test piece must remain constant. Therefore, $A_1 > A_2 > A_3$.

The curve with a solid line in Fig. 5 is obtained for engineering stresses calculated using the applied forces divided by the original cross-sectional area. The curve with a dotted line is obtained from stresses calculated using the applied forces divided by the cross-sectional area, A_2 , with the origin of this stress-strain curve located on the abscissa at the end

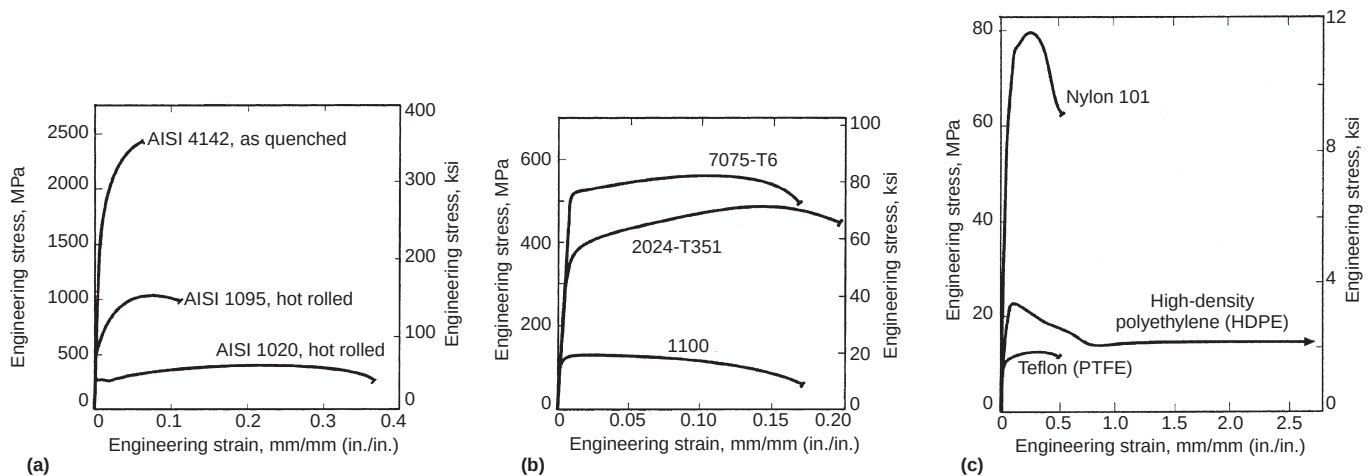


Fig. 3 Typical engineering stress-strain curves from tension tests on (a) three steels, (b) three aluminum alloys, and (c) three plastics. PTFE, polytetrafluoroethylene. Source: Ref 3

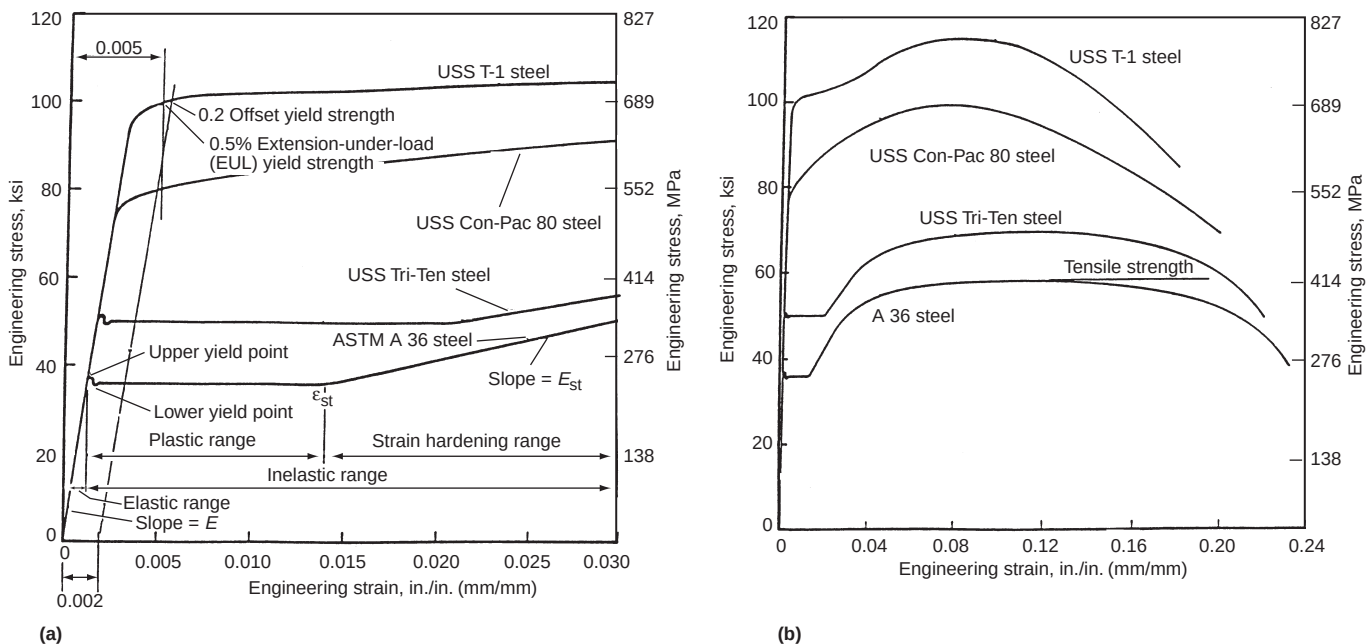


Fig. 4 Typical stress-strain curves for structural steels having specified minimum tensile properties. (a) Portions of the stress-strain curves in the yield-strength region. (b) Stress-strain curves extended through failure. Source: Ref 4

point of the first unloading line. The curve represented by the dashed line is obtained from the stresses calculated using the applied forces divided by the cross-sectional area, A_3 , with the origin of this stress-strain curve located on the abscissa at the end point of the second unloading line. This figure illustrates what happens if a test is stopped, unloaded, and restarted. It also illustrates one of the problems that can occur when testing pieces from material that has been formed into a part (or otherwise plastically strained before testing). An example is a test piece that was machined from a failed structure to determine the tensile properties. If the test piece is from a location that was subjected to tensile deformation during the failure, the properties obtained are probably not representative of the original properties of the material.

Bauschinger Effect. The other loading condition occurs when the test piece is initially loaded in compression beyond the elastic limit and then unloaded. The unload path is parallel to the initial load path but offset by the set; on reloading in tension, the elastic limit is much lower, and the shape of the stress-strain curve is significantly different. The same phenomenon occurs if the initial loading is in tension and the subsequent loading is in compression. This condition is called the **Bauschinger effect**, named for the German scientist who first described it around 1860. Again, the significance of this phenomenon is that if a test piece is machined from a location that has been subjected to plastic deformation, the stress-strain properties will be significantly different than if the material had not been so strained. This occurrence is illustrated in Fig. 6, where a machined round steel test piece was first loaded in tension to about 1% strain, unloaded, loaded in compression to about 1% strain, unloaded, and re-

loaded in tension. For this steel, the initial portion of tension and compression stress-strain curves are essentially identical.

Properties from Test Results

A number of tensile properties can be determined from the stress-strain diagram. Two of these properties, the tensile strength and the yield strength, are described in the next section of this article, "Strength Properties." In addition, total elongation (ASTM E 6), yield-point elongation (ASTM E 6), Young's modulus (ASTM E 111), and the strain-hardening exponent (ASTM E 646) are sometimes determined from the stress-strain diagram. Other tensile properties include the following:

- Poisson's ratio (ASTM E 132)
- Plastic-strain ratio (ASTM E 517)
- Elongation by manual methods (ASTM E 8)
- Reduction of area

These properties require more information than just the data pairs generating a stress-strain curve. None of these four properties can be determined from a stress-strain diagram.

Strength Properties

Tensile strength and yield strength are the most common strength properties determined in a tension test. According to ASTM E 6, tensile strength is calculated from the maximum force during a tension test that is carried to rupture divided by the original cross-sectional area of the test piece. By this definition, it is a stress value, although some product specifications de-

fine the tensile strength as the force (load) sustaining ability of the product without consideration of the cross-sectional area. Fastener specifications, for example, often refer to tensile strength as the applied force (load-carrying) capacity of a part with specific dimensions.

The yield strength refers to the stress at which a small, but measurable, amount of inelastic or plastic deformation occurs. There are three common definitions of yield strength:

- Offset yield strength
- Extension-under-load (EUL) yield strength
- Upper yield strength (or upper yield point)

An upper yield strength (upper yield point) (Fig. 7a) usually occurs with low-carbon steels and some other metal systems to a limited degree. Often, the pronounced peak of the upper yield is suppressed due to slow testing speed or nonaxial loading (i.e., bending of the test piece), metallurgical factors, or a combination of these; in this case, a curve of the type shown in Fig. 7(b) is obtained. The other two definitions of yield strength, EUL and offset, were developed for materials that do not exhibit the yield-point behavior shown in Fig. 7. Stress-strain curves without a yield point are illustrated in Fig. 4(a) for USS Con-Pac 80 and USS T-1 steels. To determine either the EUL or the offset yield strength, the stress-strain curve must be determined during the test. In computer-controlled testing systems, this curve is often stored in memory and may not be charted or displayed.

Upper yield strength (or upper yield point) can be defined as the stress at which measurable strain occurs without an increase in the stress; that is, there is a horizontal region of the

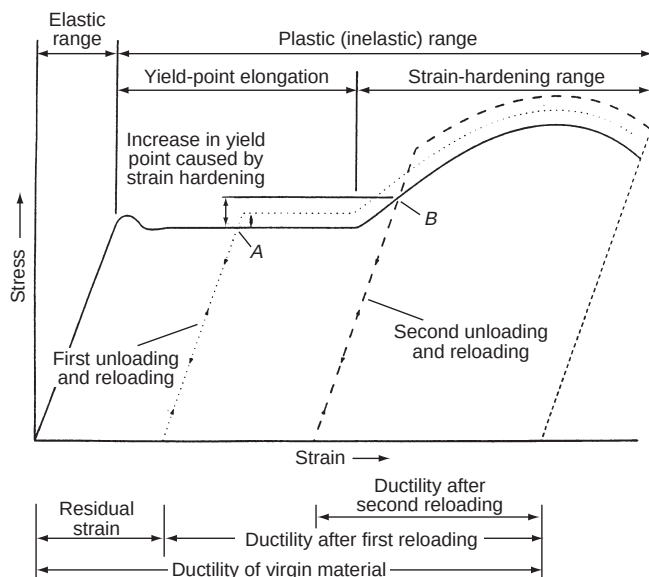


Fig. 5 Effects of prior tensile loading on tensile stress-strain behavior. Solid line, stress-strain curve based on dimensions of unstrained test piece (unloaded and reloaded twice); dotted line, stress-strain curve based on dimensions of test piece after first unloading; dashed line, stress-strain curve based on dimensions of test piece after second unloading. Note: Graph is not to scale.

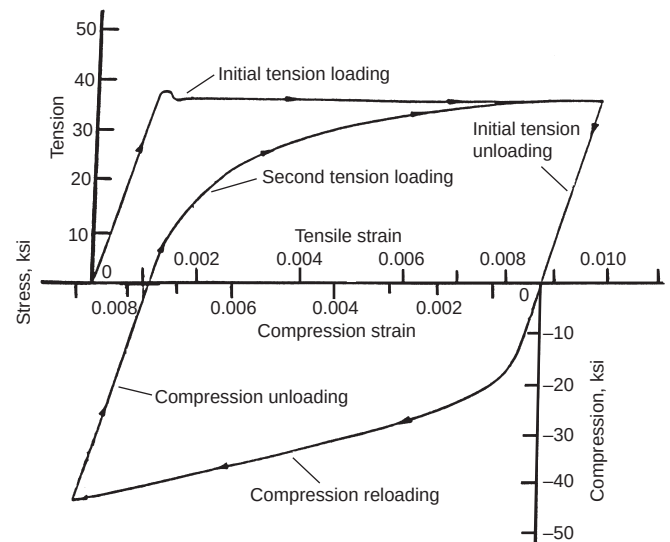


Fig. 6 Example of the Bauschinger effect and hysteresis loop in tension-compression-tension loading. This example shows initial tension loading to 1% strain, followed by compression loading to 1% strain, and then a second tension loading to 1% strain.

stress-strain curve (Fig. 7) where discontinuous yielding occurs. Before the onset of discontinuous yielding, a peak of maximum stress for yielding is typically observed (Fig. 7a). This pronounced yielding, of the type shown, is usually called yield-point elongation (YPE). This elongation is a diffusion-related phenomenon, where under certain combinations of strain rate and temperature as the material deforms, interstitial atoms are dragged along with dislocations, or dislocations can alternately break away and be repinned, with little or no increase in stress. Either or both of these actions cause serrations or discontinuous changes in a stress-strain curve, which are usually limited to the onset of yielding. This type of yield point is sometimes referred to as the upper yield strength or upper yield point. This type of yield point is usually associated with low-carbon steels, although other metal systems may exhibit yield points to some degree. For example, the stress-strain curves for A36 and USS Tri-Ten steels shown in Fig. 4(a) exhibit this behavior.

The yield point is easy to measure because the increase in strain that occurs without an increase in stress is visually apparent during the conduct of the test by observing the force-indicating system. As shown in Fig. 7, the yield

point is usually quite obvious and thus can easily be determined by observation during a tension test. It can be determined from a stress-strain curve or by the halt of the dial when the test is performed on machines that use a dial to indicate the applied force. However, when watching the movement of the dial, sometimes a minimum value, recorded during discontinuous yielding, is noted. This value is sometimes referred to as the lower yield point. When the value is ascertained without instrumentation readouts, it is often referred to as the halt-of-dial or the drop-of-beam yield point (as an average usually results from eye readings). It is almost always the upper yield point that is determined from instrument readouts.

Extension-under-load (EUL) yield strength is the stress at which a specified amount of stretch has taken place in the test piece. The EUL is determined by the use of one of the following types of apparatus:

- Autographic devices that secure stress-strain data, followed by an analysis of this data (graphically or using automated methods) to determine the stress at the specified value of extension
- Devices that indicate when the specified extension occurs so that the stress at that point may be ascertained

Graphical determination is illustrated in Fig. 8. On the stress-strain curve, the specified amount of extension, $0-m$, is measured along the strain axis from the origin of the curve and a vertical line, $m-n$, is raised to intersect the stress-strain curve. The point of intersection, r , is the EUL yield strength, and the value R is read from the stress axis. Typically, for many materials, the extension specified is 0.5%; however, other values may be specified. Therefore, when reporting the EUL, the extension also must be reported. For example, yield strength (EUL = 0.5%) = 52,500 psi is a correct way to report an EUL yield strength. The value deter-

mined by the EUL method may also be termed a yield point.

Offset yield strength is the stress that causes a specified amount of set to occur; that is, at this stress, the test piece exhibits plastic deformation (set) equal to a specific amount. To determine the offset yield strength, it is necessary to secure data (autographic or numerical) from which a stress-strain diagram may be constructed graphically or in computer memory. Figure 9 shows how to use these data; the amount of the specified offset $0-m$ is laid out on the strain axis. A line, $m-n$, parallel to the modulus of elasticity line, $0-A$, is drawn to intersect the stress-strain curve. The point of intersection, r , is the offset yield strength, and the value R is read from the stress axis. Typically, for many materials, the offset specified is 0.2%; however, other values may be specified. Therefore, when reporting the offset yield strength, the amount of the offset also must be reported; for example, "0.2 % offset yield strength = 52.8 ksi" or "yield strength (0.2% offset) = 52.8 ksi" are common formats used in reporting this information.

In Fig. 8 and 9, the initial portion of the stress-strain curve is shown in ideal terms as a straight line. Unfortunately, the initial portion of the stress-strain curve sometimes does not begin as a straight line but rather has either a concave or a convex foot (Fig. 10) (Ref 5). The shape of the initial portion of a stress-strain curve may be influenced by numerous factors such as, but not limited to, the following:

- Seating of the test piece in the grips
- Straightening of a test piece that is initially bent by residual stresses or bent by coil set
- Initial speed of testing

Generally, the aberrations in this portion of the curve should be ignored when fitting a modulus line, such as that used to determine the origin of the curve. As shown in Fig. 10, a "foot correction" may be determined by fitting a line, whether by eye or by using a computer program, to the linear portion and then extending

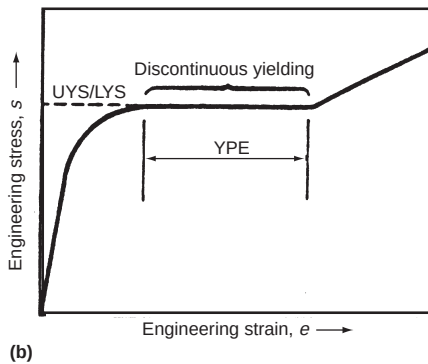
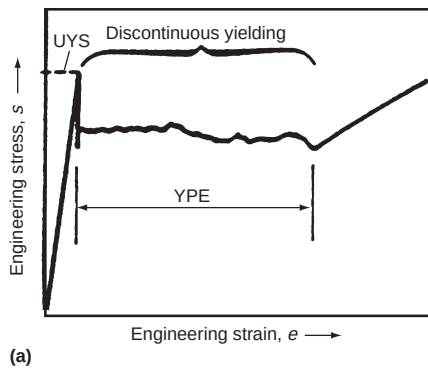


Fig. 7 Examples of stress-strain curves exhibiting pronounced yield-point behavior. Pronounced yielding, of the type shown, is usually called yield-point elongation (YPE). (a) Classic example of upper-yield-strength (UYS) behavior typically observed in low-carbon steels with a very pronounced upper yield strength. (b) General example of pronounced yielding without an upper yield strength. LYS, lower yield strength

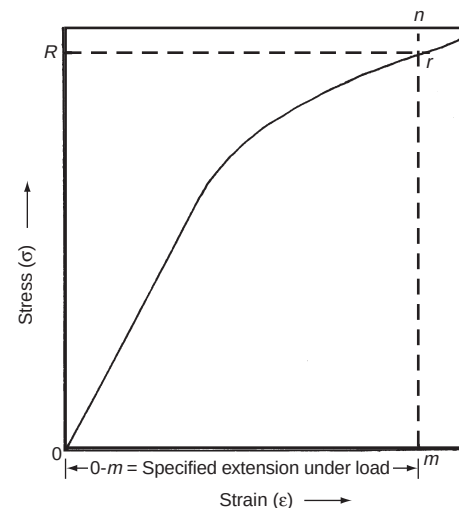


Fig. 8 Method of determining yield strength by the extension-under-load method (EUL) (adaptation of Fig. 22 in ASTM E 8)

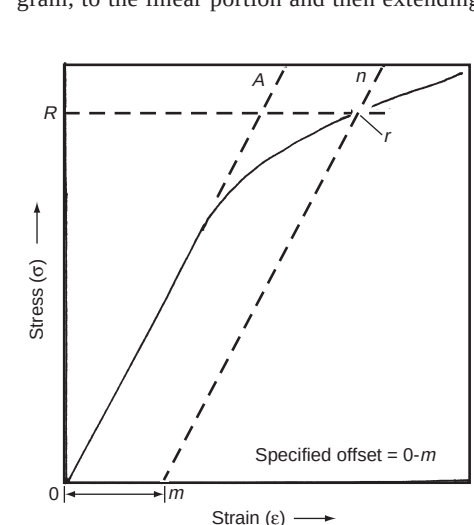


Fig. 9 Method of determining yield strength by the offset method (adaptation of Fig. 21 in ASTM E 8)

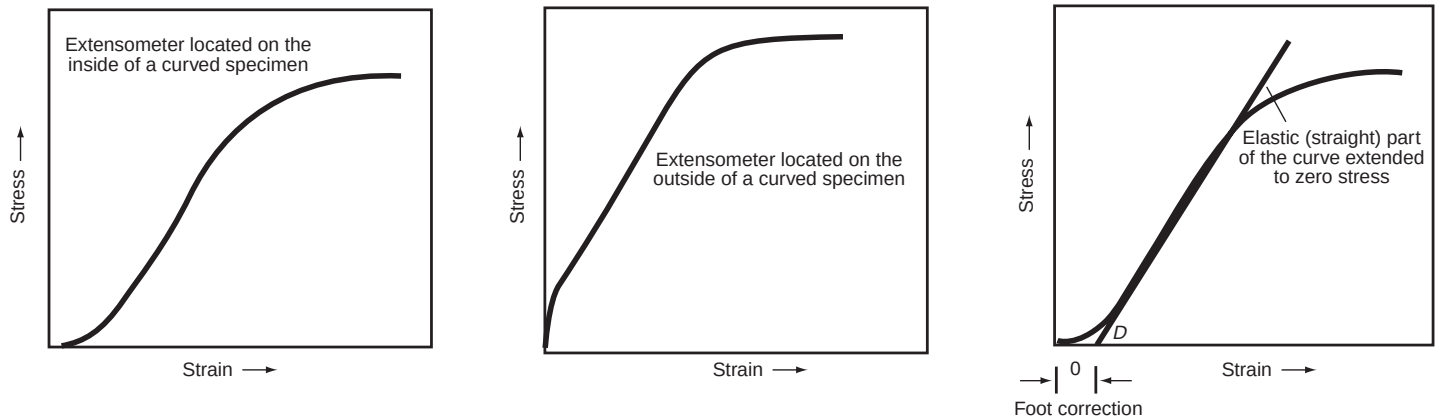


Fig. 10 Examples of stress-strain curves requiring foot correction. Point *D* is the point where the extension of the straight (elastic) part diverges from the stress-strain curve. Source: Ref 5

this line back to the abscissa, which becomes point 0 in Fig. 8 and 9. As a rule of thumb, Point *D* in Fig. 10 should be less than one-half the specified yield point or yield strength.

Tangent or Chord Moduli. For materials that do not have a linear relationship between stress and strain, even at very low stresses, the offset yield is meaningless without defining how to determine the modulus of elasticity. Often, a chord modulus or a tangent modulus is specified. A chord modulus is the slope of a chord between any two specified points on the stress-strain curve, usually below the elastic limit. A tangent modulus is the slope of the stress-strain curve at a specified value of stress or of strain. Chord and tangent moduli are illustrated in Fig. 11. Another technique that has been used is sketched in Fig. 12. The test piece is stressed to approximately the yield strength, unloaded to about 10% of this value, and reloaded. As previously discussed, the unloading line will be parallel to what would have been the initial modulus line, and the reloading line will coincide with the unloading line (assuming no hysteresis in any of the system components). The slope of this line is transferred to the initial loading line, and the offset is determined as before. The stress or strain at which the test piece is unloaded usually is not important. This technique

is specified in the ISO standard for the tension test of metallic materials, ISO 6892.

Yield-strength-property values generally depend on the definition being used. As shown in Fig. 4(a) for the USS Con-Pac steel, the EUL yield is greater than the offset yield, but for the USS T-1 steel (Fig. 4a), the opposite is true. The amount of the difference between the two values is dependent upon the slope of the stress-strain curve between the two intersections. When the stress-strain data pairs are sampled by a computer, and a yield spike or peak of the type shown in Fig. 7(a) occurs, the EUL and the offset yield strength will probably be less than the upper yield point and will probably differ because the *m-n* lines of Fig. 8 and 9 will intersect at different points in the region of discontinuous yielding.

Ductility

Ductility is the ability of a material to deform plastically without fracturing. Figure 13 is a sketch of a test piece with a circular cross section that has been pulled to fracture. As indicated in this sketch, the test piece elongates during the tension test and correspondingly reduces in cross-sectional area. The two mea-

sures of the ductility of a material are the amount of elongation and reduction in area that occurs during a tension test.

Elongation, as previously noted, is defined in ASTM E 6 as the increase in the gage length of a test piece subjected to a tension force, divided by the original gage length on the test piece. Elongation usually is expressed as a percentage of the original gage length. ASTM E 6 further indicates the following:

- The increase in gage length may be determined either at or after fracture, as specified for the material under test.
- The gage length shall be stated when reporting values of elongation.
- Elongation is affected by test-piece geometry (gage length, width, and thickness of the gage section and of adjacent regions) and test procedure variables, such as alignment and speed of pulling.

The manual measurement of elongation on a tension test piece can be done with the aid of gage marks applied to the unstrained reduced section. After the test, the amount of stretch between gage marks is measured with an appropriate device. The use of the term elongation in this instance refers to the total amount of

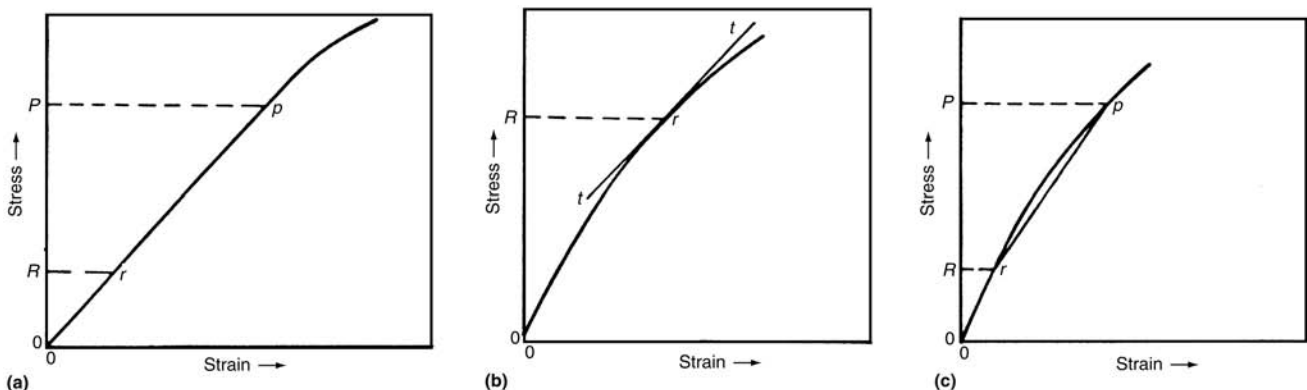


Fig. 11 Stress-strain curves showing straight lines corresponding to (a) Young's modulus between stress, *P*, below proportional limit and *R*, or preload; (b) tangent modulus at any stress, *R*; and (c) chord modulus between any two stresses, *P* and *R*. Source: Ref 6

stretch or extension. Elongation, in the sense of nominal engineering strain, e , is the value of gage extension divided by the original distance between the gage marks. Strain elongation is usually expressed as a percentage, where the nominal engineering strain is multiplied by 100 to obtain a percent value; that is:

$$e, \% = \left[\frac{(\text{final gage length} - \text{original gage length})}{\text{original gage length}} \right] \times 100$$

The final gage length at the completion of the test may be determined in two ways. Historically, it was determined manually by carefully fitting the two ends of the fractured test piece together (Fig. 13) and measuring the distance between the gage marks. However, some modern computer-controlled testing systems obtain data from an extensometer that is left on the test piece through fracture. In this case, the computer may be programmed to report the elongation as the last strain value obtained prior to some event, perhaps the point at which the applied force drops to 90% of the maximum value recorded. There has been no general agreement about what event should be the trigger, and users and machine manufacturers find that different events may be appropriate for dif-

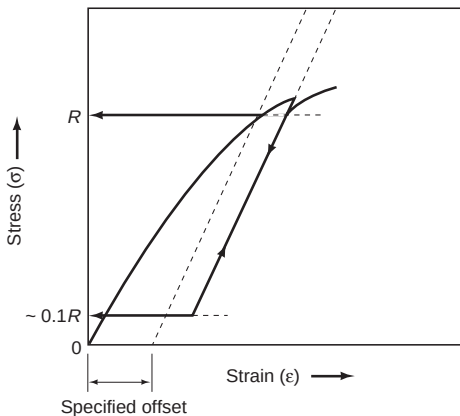


Fig. 12 Alternate technique for establishing Young's modulus for a material without an initial linear portion

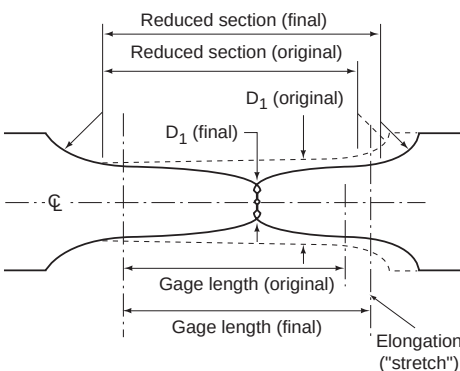


Fig. 13 Sketch of fractured, round tension test piece. Dashed lines show original shape. Strain = elongation/gage length

ferent materials (although some consensus has been reached, see ASTM E 8-99). The elongation values determined by these two methods are not the same; in general, the result obtained by the manual method is a couple of percent larger and is more variable because the test-piece ends do not fit together perfectly. It is strongly recommended that when disagreements arise about elongation results, agreement should be reached on which method will be used prior to any further testing.

Test methods often specify special conditions that must be followed when a product specification specifies elongation values that are small, or when the expected elongation values are small. For example, ASTM E 8 defines small as 3% or less.

Effect of Gage Length and Necking. Figure 14 (Ref 7) shows the effect of gage length on elongation values. Gage length is very important; however, as the gage length becomes quite large, the elongation tends to be independent of the gage length. The gage length must be specified prior to the test, and it must be shown in the data record for the test.

Figures 13 and 14 also illustrate considerable localized deformation in the vicinity of the fracture. This region of local deformation is often called a neck, and the occurrence of this deformation is termed necking. Necking occurs as the force begins to drop after the maximum force has been reached on the stress-strain curve. Up to the point at which the maximum force occurs, the strain is uniform along the gage length; that is, the strain is independent of

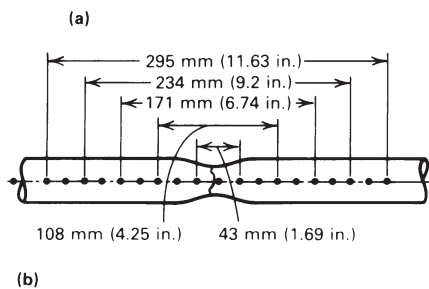
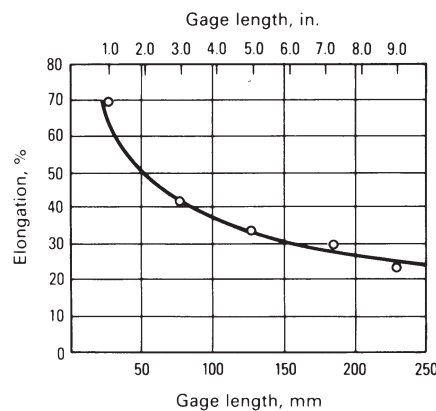


Fig. 14 Effect of gage length on the percent elongation. (a) Elongation, %, as a function of gage length for a fractured tension test piece. (b) Distribution of elongation along a fractured tension test piece. Original spacing between gage marks, 12.5 mm (0.5 in.). Source: Ref 7

the gage length. However, once necking begins, the gage length becomes very important. When the gage length is short, this localized deformation becomes the principal portion of measured elongation. For long gage lengths, the localized deformation is a much smaller portion of the total. For this reason, when elongation values are reported, the gage length must also be reported, for example, elongation = 25% (50 mm, or 2.00 in., gage length).

Effect of Test-Piece Dimensions. Test-piece dimensions also have a significant effect on elongation measurements. Experimental work has verified the general applicability of the following equation:

$$e = e_0 (L/A^{1/2})^{-a}$$

where e_0 is the specific elongation constant; $L/A^{1/2}$ the slimmness ratio, K , of gage length, L , and cross-sectional areas, A ; and a is another material constant. This equation is known as the Bertella-Oliver equation, and it may be transformed into logarithmic form and plotted as shown in Fig. 15. In one study, quadruplet sets of machined circular test pieces (four different diameters ranging from 0.125 to 0.750 in.) and rectangular test pieces ($\frac{1}{2}$ in. wide with three thicknesses and $1\frac{1}{2}$ in. wide with three thicknesses) were machined from a single plate. Multiple gage lengths were scribed on each test piece to produce a total of 40 slimmness

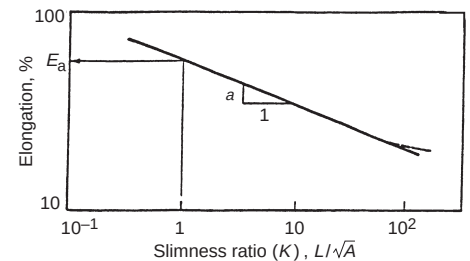


Fig. 15 Graphical form of the Bertella-Oliver equation.

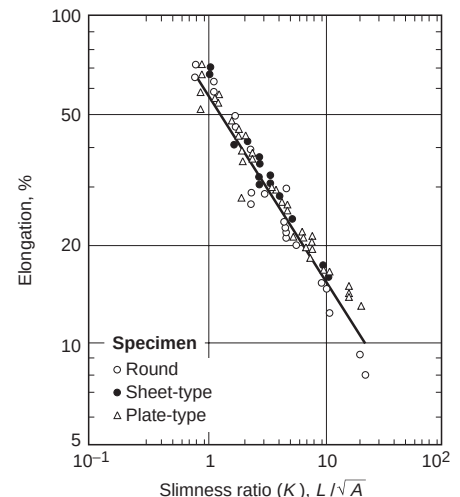


Fig. 16 Graphical form of the Bertella-Oliver equation showing actual data

ratios. The results of this study, for one of the grades of steel tested, are shown in Fig. 16.

In order to compare elongation values of test pieces with different slimmness ratios, it is necessary only to determine the value of the material constant, a . This calculation can be made by testing the same material with two different geometries (or the same geometry with different gage lengths) with different slimmness ratios, K_1 and K_2 , where

$$e_0 = e_1 / K_1^{-a} = e_2 / K_2^{-a}$$

solving for a , then:

$$(K_2 / K_1)^{-a} = e_2 / e_1$$

or:

$$-a = \frac{\ln(e_2 / e_1)}{\ln(K_2 / K_1)} \quad (\text{Eq 1})$$

$$-a = \frac{\ln(e_2) - \ln(e_1)}{\ln(K_2) - \ln(K_1)} \quad (\text{Eq 2})$$

The values of the e_0 and a parameters depend on the material composition, the strength, and the material condition and are determined empirically with a best-fit line plot around data points. Reference 8 specifies "value $a = 0.4$ for carbon, carbon-manganese, molybdenum, and chromium-molybdenum steels within the tensile strength range of 275 to 585 MPa (40 to 85 ksi) and in the hot-rolled, in the hot-rolled and normalized, or in the annealed condition, with or without tempering. Materials that have been cold reduced require the use of a different value for a , and an appropriate value is not suggested." Reference 8 uses a value of $a = 0.127$ for annealed, austenitic stainless steels. However, Ref 8 states that "these conversions shall not be used where the width-to-thickness ratio, w/t , of the test piece exceeds 20." ISO 2566/1 (Ref 9) contains similar statements. In addition to the limit of $(w/t) < 20$, Ref 9 also specifies that the slimmness ratio shall be less than 25.

Some tension-test specifications do not contain standard test-piece geometries but require that the slimmness ratio be either 5.65 or 11.3. For a round test piece, a slimmness ratio of 5.65 produces a 5-to-1 relation between the diameter and the gage length, and a slimmness ratio of 4.51 produces a 4-to-1 relation between the diameter and gage length (which is that of the test piece in ASTM E 8).

Reduction of area is another measure of the ductility of metal. As a test piece is stretched, the cross-sectional area decreases, and as long as the stretch is uniform, the reduction of area is proportional to the amount of stretch or extension. However, once necking begins to occur, proportionality is no longer valid.

According to ASTM E 6, reduction of area is defined as "the difference between the original cross-sectional area of a tension test piece and the area of its smallest cross section." Reduction of area is usually expressed as a percentage of the original cross-sectional area of the test

piece. The smallest final cross section may be measured at or after fracture as specified for the material under test. The reduction of area (RA) is almost always expressed as a percentage:

$$Ra, \% = \left[\frac{(\text{original area} - \text{final area})}{\text{original area}} \right] \times 100$$

Reduction of area is customarily measured only on test pieces with an initial circular cross section because the shape of the reduced area remains circular or nearly circular throughout the test for such test pieces. With rectangular test pieces, in contrast, the corners prevent uniform flow from occurring, and consequently, after fracture, the shape of the reduced area is not rectangular (Fig. 17). Although a number of expressions have been used in an attempt to describe the way to determine the reduced area, none has received general agreement. Thus, if a test specification requires the measurement of the reduction of area of a test piece that is not circular, the method of determining the reduced area should be agreed to prior to performing the test.

General Procedures

Numerous groups have developed standard methods for conducting the tension test. In the United States, standards published by ASTM are commonly used to define tension-test procedures and parameters. Of the various ASTM standards related to tension tests (for example, those listed in "Selected References" at the end of this article), the most common method for tension testing of metallic materials is ASTM E 8 "Standard Test Methods for Tension Testing of Metallic Materials" (or the version using metric units, ASTM E 8M). Standard methods for conducting the tension test are also available from other standards organizations, such as the Japanese Industrial Standards (JIS), the Deutsche Institut für Normung (DIN), and the International Organization for Standardization (ISO). Other domestic technical groups in the United States have developed standards, but in general, these are based on ASTM E 8.

With the increasing internationalization of trade, methods developed by other national standards organizations (such as JIS, DIN, or ISO standards) are increasingly being used in the United States. Although most tension-test standards address the same concerns, they differ in the values assigned to variables. Thus, a tension test performed in accordance with ASTM E 8 will not necessarily have been conducted in accordance with ISO 6892 or JIS Z2241, and so on, and vice versa. Therefore, it is necessary to specify the applicable testing standard for any test results or mechanical property data.

Unless specifically indicated otherwise, the values of all variables discussed hereafter are those related to ASTM E 8 "Standard Test Methods for Tension Testing of Metallic Materials." A flow diagram of the steps involved

when a tension test is conducted in accordance with ASTM E 8 is shown in Fig. 18. The test consists of three distinct parts:

- Test-piece preparation, geometry, and material condition
- Test setup and equipment
- Test

The Test Piece

The test piece is one of two basic types. Either it is a full cross section of the product form, or it is a small portion that has been machined to specific dimensions. Full-section test pieces consist of a part of the test unit as it is fabricated. Examples of full-section test pieces include bars, wires, and hot-rolled or extruded angles cut to a suitable length and then gripped at the ends and tested. In contrast, a machined test piece is a representative sample, such as one of the following:

- Test piece machined from a rough specimen taken from a coil or plate
- Test piece machined from a bar with dimensions that preclude testing a full-section test piece because a full-section test piece exceeds the capacity of the grips or the force capacity of the available testing machine or both
- Test piece machined from material of great monetary or technical value

In these cases, representative samples of the material must be obtained for testing. The descriptions of the tension test in this article proceed from the point that a rough specimen (Fig. 19) has been obtained. That is, the rough specimen has been selected based on some criteria, usually a material specification or a test order issued for a specific reason.

In this article, the term *test piece* is used for what is often called a specimen. This terminology is based on the convention established by ISO Technical Committee 17, Steel in ISO 377-1, "Selection and Preparation of Samples and Test Pieces of Wrought Steel," where terms for a test unit, a sample product, sample, rough specimen, and test piece are defined as follows:

- **Test unit:** The quantity specified in an order that requires testing (for example, 10 tons of $\frac{3}{4}$ in. bars in random lengths)
- **Sample product:** Item (in the previous example, a single bar) selected from a test unit for the purpose of obtaining the test pieces

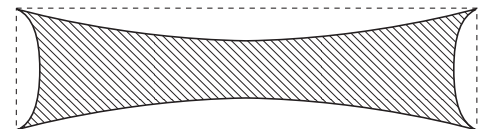


Fig. 17 Sketch of end view of rectangular test piece after fracture showing constraint at corners indicating the difficulty of determining reduced area

- **Sample:** A sufficient quantity of material taken from the sample product for the purpose of producing one or more test pieces. In some cases, the sample may be the sample product itself (i.e., a 2 ft length of the sample product).
- **Rough specimen:** Part of the sample having undergone mechanical treatment, followed by heat treatment where appropriate, for the purpose of producing test pieces; in the example, the sample is the rough specimen.
- **Test piece:** Part of the sample or rough specimen, with specified dimensions, machined or unmachined, brought to the required condition for submission to a given test. If a testing machine with sufficient force capacity is available, the test piece may be the rough specimen; if sufficient capacity is not available, or for another reason, the test piece may be machined from the rough specimen to dimensions specified by a standard.

These terms are shown graphically in Fig. 19. As can be seen, the test piece, or what is commonly called a specimen, is a very small part of the entire test unit.

Description of Test Material

Test-Piece Orientation. Orientation and location of a test material from a product can influence measured tensile properties. Although modern metal-working practices, such as cross rolling, have tended to reduce the magnitude of the variations in the tensile properties, it must not be neglected when locating the test piece within the specimen or the sample.

Because most materials are not isotropic, test-piece orientation is defined with respect to a set of axes as shown in Fig. 20. These terms for the orientation of the test-piece axes in Fig. 20 are based on the convention used by ASTM E 8 "Fatigue and Fracture." This scheme is identical to that used by the ISO Technical Committee 164 "Mechanical Testing," although the *L*, *T*, and *S* axes are referred to as the *X*, *Y*, and *Z* axes, respectively, in the ISO documents.

When a test is being performed to determine conformance to a product standard, the product standard must state the proper orientation of the test piece with regard to the axis of prior working, (e.g., the rolling direction of a flat product). Because alloy systems behave differently, no general rule of thumb can be stated on how prior working may affect the directionality of properties. As can be seen in Table 1, the longitudinal strengths of steel are generally somewhat less than the transverse strength. However, for aluminum alloys, the opposite is generally true.

Many standards, such as ASTM A 370, E 8, and B 557, provide guidance in the selection of test-piece orientation relative to the rolling direction of the plate or the major forming axes of other types of products and in the selection of specimen and test-piece location relative to the surface of the product. Orientation is also important when characterizing the directionality

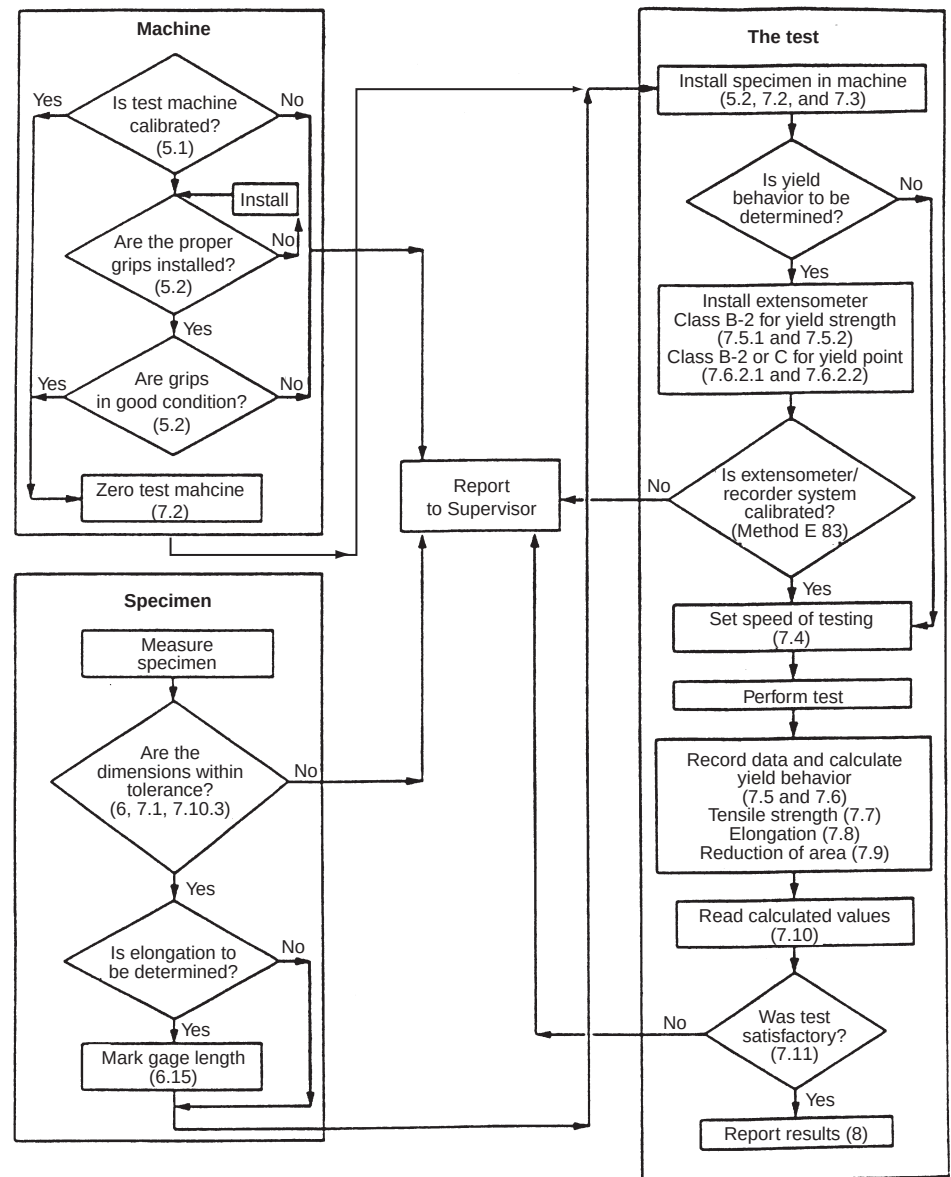


Fig. 18 General flow chart of the tension test per procedures in ASTM E 8. Relevant paragraph numbers from ASTM E 8 are shown in parentheses.

of properties that often develops in the microstructure of materials during processing. For example, some causes of directionality include the fibering of inclusions in steels, the formation of crystallographic textures in most metals and alloys, and the alignment of molecular chains in polymers.

The location from which a test material is taken from the initial product form is important because the manner in which a material is processed influences the uniformity of microstructure along the length of the product as well as through its thickness properties. For example, the properties of metal cut from castings are influenced by the rate of cooling and by shrinkage stresses at changes in section. Generally, test pieces taken from near the surface of iron castings are stronger. To standardize test results relative to location, ASTM A 370 rec-

ommends that tension test pieces be taken from midway between the surface and the center of round, square, hexagon, or octagonal bars. ASTM E 8 recommends that test pieces be taken from the thickest part of a forging from which a test coupon can be obtained, from a prolongation of the forging, or in some cases, from separately forged coupons representative of the forging.

Test-Piece Geometry

As previously noted, the item being tested may be either the full cross section of the item or a portion of the item that has been machined to specific dimensions. This article focuses on tension testing with test pieces that are machined from rough samples. Component testing is discussed in more detail in the article

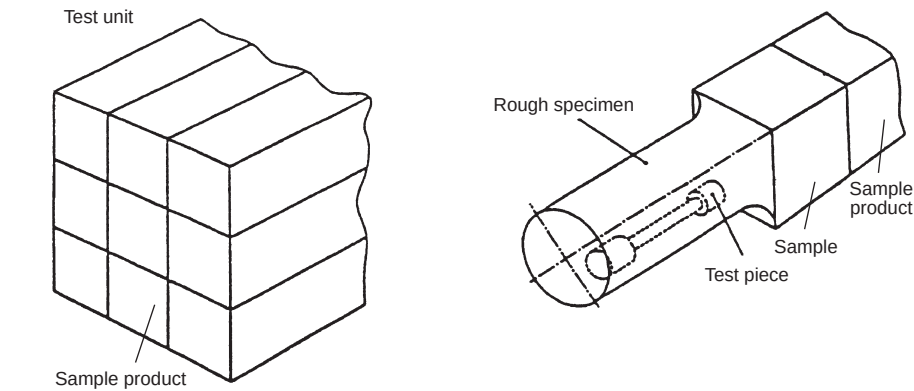


Fig. 19 Illustration of ISO terminology used to differentiate between sample, specimen, and test piece (see text for definitions of test unit, sample product, sample, rough specimen, and test piece). As an example, a test unit may be a 250-ton heat of steel that has been rolled into a single thickness of plate. The sample product is thus one plate from which a single test piece is obtained.

“Mechanical Testing of Fiber Reinforced Composites” in this Volume.

Test-piece geometry is often influenced by product form. For example, only test pieces with rectangular cross sections can be obtained from sheet products. Test pieces taken from thick plate may have either flat (plate-type) or round cross sections. Most tension-test specifications show machined test pieces with either circular cross sections or rectangular cross sections. Nomenclature for the various sections of a machined test piece are shown in Fig. 21. Most tension-test specifications present a set of dimensions, for each cross-section type, that are standard, as well as additional sets of dimensions for alternative test pieces. In general,

the standard dimensions published by ASTM, ISO, JIS, and DIN are similar, but they are not identical.

Gage lengths and standard dimensions for machined test pieces specified in ASTM E 8 are shown in Fig. 22 for rectangular and round test pieces. From this figure, it can be seen that the gage length is proportionally four times (4 to 1) the diameter (or width) of the test piece for the standard machined round test pieces and the sheet-type, rectangular test pieces. The length of the reduced section is also a *minimum* of 4½ times the diameter (or width) of these

test-piece types. These relationships do not apply to plate-type rectangular test pieces.

Many specifications outside the United States require that the gage length of a test piece be a fixed ratio of the square root of the cross-sectional area, that is:

Gage length = constant x (cross-sectional area)^{1/2}

The value of this constant is often specified as 5.65 or 11.3 and applies to both round and rectangular test pieces. For machined round test pieces, a value of 5.65 results in a 5-to-1 relationship between the gage length and the diameter.

Many tension-test specifications permit a slight taper toward the center of the reduced section of machined test pieces so that the minimum cross section occurs at the center of the gage length and thereby tends to cause fracture to occur at the middle of the gage length. ASTM E 8-99 specifies that this taper cannot exceed 1% and requires that the taper is the same on both sides of the midlength.

When test pieces are machined, it is important that the longitudinal centerline of the reduced section be coincident with the longitudinal centerlines of the grip ends. In addition, for the rectangular test pieces, it is essential that the centers of the transition radii at each end of the reduced section are on common lines that are perpendicular to the longitudinal centerline.

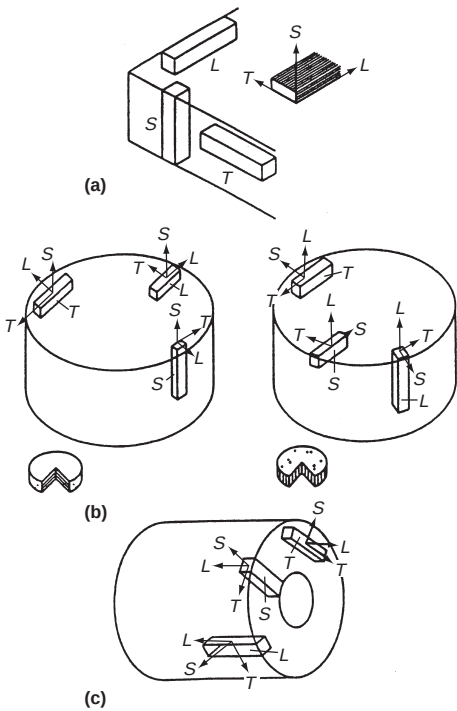


Fig. 20 System for identifying the axes of test-piece orientation in various product forms. (a) Flat-rolled products. (b) Cylindrical sections. (c) Tubular products

Table 1 Effect of test-piece orientation on tensile properties

Orientation	Yield strength, ksi	Tensile strength, ksi	Elongation in 50 mm (2 in.), %	Reduction of area, %
ASTM A 572, Grade 50 (¼ in. thick plate, low sulfur level)				
Longitudinal	58.8	84.0	27.0	70.2
Transverse	59.8	85.2	28.0	69.0
ASTM A 656, Grade 80 (¼ in. thick plate, low sulfur level + controlled rolled)				
Longitudinal	81.0	102.3	25.8	71.2
Transverse	86.9	107.9	24.5	67.1
ASTM A 5414 (¼ in. thick plate, low sulfur level)				
Longitudinal	114.6	121.1	19.8	70.6
Transverse	116.3	122.2	19.5	69.9

Source: Courtesy of Francis J. Marsh

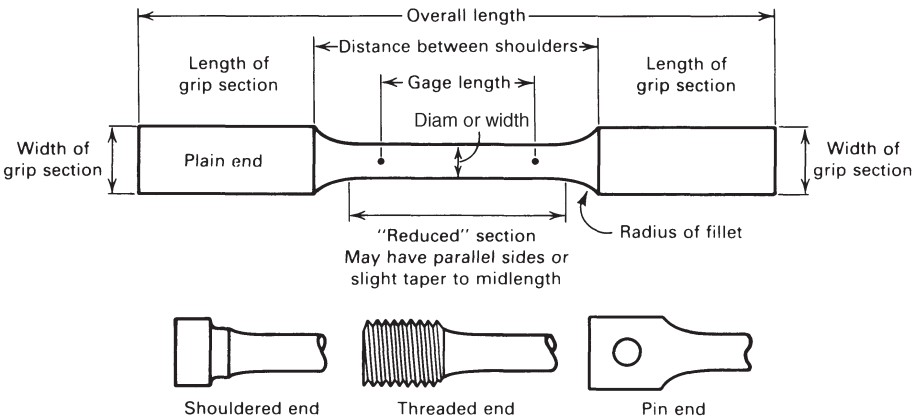


Fig. 21 Nomenclature for a typical tension test piece

If any of these requirements is violated, bending will occur, which may affect test results.

The transition radii between the reduced section and the grip ends can be critical for test pieces from materials with very high strength or with very little ductility or both. This is discussed more fully in the section "Effect of Strain Concentrations" in this article.

Measurement of Initial Test-Piece Dimensions. Machined test pieces are expected to meet size specifications, but to ensure dimensional accuracy, each test piece should be measured prior to testing. Gage length, fillet radius, and cross-sectional dimensions are measured easily. Cylindrical test pieces should be measured for concentricity. Maintaining acceptable concentricity is extremely important in minimizing unintended bending stresses on materials in a brittle state.

Measurement of Cross-Sectional Dimensions. The test pieces must be measured to determine whether they meet the requirements of the test method. Test-piece measurements must also determine the initial cross-sectional area when it is compared against the final cross section after testing as a measure of ductility.

The precision with which these measurements are made is based on the requirements of the test method, or if none are given, on good engineering judgment. Specified requirements of ASTM E 8 are summarized as follows:

- For referee testing of test pieces under $\frac{3}{16}$ in. in their least dimension, the dimensions should be measured where the least cross-sectional area is found.
- For cross sectional dimensions of 0.200 in. or more, cross-sectional dimensions should be measured and recorded to the nearest 0.001 in.
- For cross sectional dimensions from 0.100 in. but less than 0.200 in., cross-sectional dimensions should be measured and recorded to the nearest 0.0005 in.
- For cross sectional dimensions from 0.020 in. but less than 0.100 in., cross-sectional dimensions should be measured and recorded to the nearest 0.0001 in.
- When practical, for cross-sectional dimensions less than 0.020 in., cross-sectional dimensions should be measured to the nearest 1%, but in all cases, to at least the nearest 0.0001 in.

ASTM E 8 goes on to state how to determine the cross-sectional area of a test piece that has a nonsymmetrical cross section using the weight and density. When measuring dimensions of the test piece, ASTM E 8 makes no distinction between the shape of the cross section for standard test pieces.

Measurement of the Initial Gage Length. ASTM E 8 assumes that the initial gage length is within specified tolerance; therefore, it is necessary only to verify that the gage length of the test piece is within the tolerance.

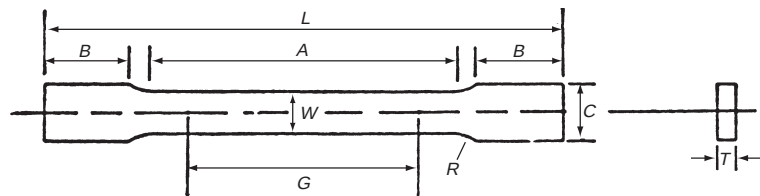
Marking Gage Length. As shown in the flow diagram in Fig. 18, measurement of elongation

requires marking the gage length of the test piece. The gage marks should be placed on the test piece in a manner so that when fracture occurs, the fracture will be located within the center one-third of the gage length (or within the center one-third of one of several sets of gage-length marks). For a test piece machined with a reduced-section length that is the minimum specified by ASTM E 8 and with a gage length equal to the maximum allowed for that geometry, a single set of marks is usually sufficient. However, multiple sets of gage lengths

must be applied to the test piece to ensure that one set spans the fracture under any of the following conditions:

- Testing full-section test pieces
- Testing pieces with reduced sections significantly longer than the minimum
- Test requirements specify a gage length that is significantly shorter than the reduced section

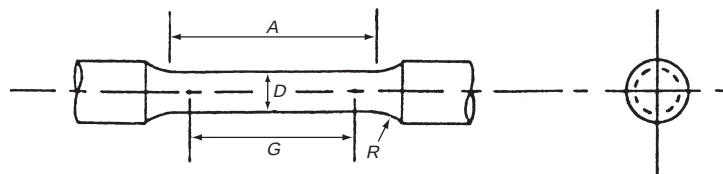
For example, some product specifications require that the elongation be measured over a 2 in.



	Standard specimens, in.		Subsize specimen, $\frac{1}{4}$ in. wide, in.
	Plate type, $1\frac{1}{2}$ in. wide	Sheet type, $\frac{1}{2}$ in. wide	
G, gage length(a)(b)	8.00 ± 0.01	2.00 ± 0.005	1.000 ± 0.003
W, width(c)(d)	$1\frac{1}{2} + \frac{1}{16} - \frac{1}{4}$	0.500 ± 0.010	0.250 ± 0.005
T, thickness(e)	$< \frac{3}{16}$	$0.005 \leq T \leq \frac{3}{4}$	$0.005 \leq T \leq \frac{1}{4}$
R, radius of fillet, min(f)	1	$\frac{1}{2}$	$\frac{1}{4}$
L, overall length, min(b)(g)	18	8	4
A, length of reduced section, min	9	$2\frac{1}{4}$	$1\frac{1}{4}$
B, length of grip section, min(h)	3	2	$1\frac{1}{4}$
C, width of grip section, approximate(d)(i)	2	$\frac{3}{4}$	$\frac{3}{8}$

Note: (a) For the $1\frac{1}{2}$ in. wide specimen, punch marks for measuring elongation after fracture shall be made on the flat or on the edge of the specimen and within the reduced section. Either a set of nine or more punch marks 1 in. apart or one or more pairs of punch marks 8 in. apart may be used. (b) When elongation measurements of $1\frac{1}{2}$ in. wide specimens are not required, a minimum length of reduced section (A) of $2\frac{1}{4}$ in. may be used with all other dimensions similar to those of the plate-type specimen. (c) For the three sizes of specimens, the ends of the reduced section shall not differ in width by more than 0.004, 0.002, or 0.001 in., respectively. Also, there may be a gradual decrease in width from the ends to the center, but the width at each end shall not be more than 0.015, 0.005, or 0.003 in., respectively, larger than the width at the center. (d) For each of the three sizes of specimens, narrower widths (W and C) may be used when necessary. In such cases the width of the reduced section should be as large as the width of the material being tested permits; however, unless stated specifically, the requirements for elongation in a product specification shall not apply when these narrower specimens are used. (e) The dimension T is the thickness of the test specimen as provided for in the applicable material specifications. Minimum thickness of $1\frac{1}{2}$ in. wide specimens shall be $\frac{3}{16}$ in. Maximum thickness of $\frac{1}{2}$ in. and $\frac{1}{4}$ in. wide specimens shall be $\frac{3}{4}$ in. and $\frac{1}{4}$ in., respectively. (f) For the $1\frac{1}{2}$ in. wide specimen, a $\frac{1}{2}$ in. minimum radius at the ends of the reduced section is permitted for steel specimens under 100,000 psi in tensile strength when a profile cutter is used to machine the reduced section. (g) To aid in obtaining axial force application during testing of $\frac{1}{4}$ in. wide specimens, the overall length should be as large as the material will permit, up to 8.00 in. (h) It is desirable, if possible, to make the length of the grip section large enough to allow the specimen to extend into the grips a distance equal to two-thirds or more of the length of the grips. If the thickness of $\frac{1}{2}$ in. wide specimens is over $\frac{3}{8}$ in., longer grips and correspondingly longer grip sections of the specimen may be necessary to prevent failure in the grip section. (i) For the three sizes of specimens, the ends of the specimen shall be symmetrical in width with the center line of the reduced section within 0.10, 0.05, and 0.005 in., respectively. However, for referee testing and when required by product specifications, the ends of the $\frac{1}{4}$ in. wide specimen shall be symmetrical within 0.01 in.

(a)



	Standard specimen, in., at nominal diameter:		Small-size specimens proportional to standard, in., at nominal diameter:		
	0.500	0.350	0.250	0.160	0.113
G, gage length	2.000 ± 0.005	1.400 ± 0.005	1.000 ± 0.005	0.640 ± 0.005	0.450 ± 0.005
D, diameter(a)	0.500 ± 0.010	0.350 ± 0.007	0.250 ± 0.005	0.160 ± 0.003	0.113 ± 0.002
R, radius of fillet, min	$\frac{3}{8}$	$\frac{1}{4}$	$\frac{3}{16}$	$\frac{7}{32}$	$\frac{1}{8}$
A, length of reduced section, min(b)	$2\frac{1}{4}$	$1\frac{1}{4}$	$1\frac{1}{4}$	$\frac{3}{4}$	$\frac{5}{8}$

Note: (a) The reduced section may have a gradual taper from the ends toward the center, with the ends not more than 1% larger in diameter than the center (controlling dimension). (b) If desired, the length of the reduced section may be increased to accommodate an extensometer of any convenient gage length. Reference marks for the measurement of elongation should, nevertheless, be spaced at the indicated gage length.

(b)

Fig. 22 Examples of tension test pieces per ASTM E 8. (a) Rectangular (flat) test pieces. (b) Round test-piece

gage length using the machined plate-type test piece with a 9 in. reduced section (Fig. 22a). In this case, it is recommended that a staggered series of marks (either in increments of 1 in. when testing to ASTM E 8 or in increments of 25.0 mm when testing to ASTM E 8M) be placed on the test piece such that, after fracture, the elongation can be measured using the set that best meets the center-third criteria. Many tension-test methods permit a retest when the elongation is less than the minimum specified by a product specification if the fracture occurred outside the center third of the gage length. When testing full-section test pieces and determining elongation, it is important that the distance between the grips be greater than the specified gage length unless otherwise specified. As a rule of thumb, the distance between grips should be equal to at least the gage length plus twice the minimum dimension of the cross section.

The gage marks may be marks made with a center punch, or may be lines scribed using a sharp, pointed tool, such as a machinist's scribe (or any other means that will establish the gage length within the tolerance permitted by the test method). If scribed lines are used, a broad line or band may first be drawn along the length of the test piece using machinist's layout ink (or a similar substance), and the gage marks are made on this line. This practice is especially helpful to improve visibility of scribed gage marks after fracture. If punched marks are used, a circle around each mark or other indication made by ink may help improve visibility after fracture. Care must be taken to ensure that the gage marks, especially those made using a punch, are not deep enough to become stress raisers, which could cause the fracture to occur through them. This precaution is especially important when testing materials with high strength and low ductility.

Notched Test Pieces. Tension test pieces are sometimes intentionally notched in the center of the gage length (Fig. 23). ASTM E 338 and E 602 describe procedures for testing

notched test pieces. Results obtained using notched test pieces are useful for evaluating the response of a material to a localized stress concentration. Detailed information on the notch tensile test and a discussion of the related material characteristics (notch sensitivity and notch strength) can be found in the article "Mechanical Behavior Under Tensile and Compressive Loads" in this Volume. The effect of stress (or strain) concentrations is also discussed in the section "Effect of Strain Concentrations" in this article.

Surface Finish and Condition. The finish of machined surfaces usually is not specified in generic test methods (that is, a method that is not written for a specific item or material) because the effect of finish differs for different materials. For example, test pieces from materials that are not high strength or that are ductile are usually insensitive to surface finish effects. However, if surface finish in the gage length of a tensile test piece is extremely poor (with machine tool marks deep enough to act as stress-concentrating notches, for example), test results may exhibit a tendency toward decreased and variable strength and ductility.

It is good practice to examine the test piece surface for deep scratches, gouges, edge tears, or shear burrs. These discontinuities may sometimes be minimized or removed by polishing or, if necessary, by further machining; however, dimensional requirements often may no longer be met after additional machining or polishing. In all cases, the reduced sections of machined test pieces must be free of detrimental characteristics, such as cold work, chatter marks, grooves, gouges, burrs, and so on. Unless one or more of these characteristics is typical of the product being tested, an unmachined test piece must also be free of these characteristics in the portion of the test piece that is between the gripping devices. When rectangular test pieces are prepared from thin-gage sheet material by shearing (punching) using a die the shape of the test piece, ASTM E 8 states that the sides of the reduced section may need to be further ma-

chined to remove the cold work and shear burrs that occur when the test piece is sheared from the rough specimen. This method is impractical for material less than 0.38 mm (0.015 in.) thick. Burrs on test pieces can be virtually eliminated if punch-to-die clearances are minimized.

Test Setup

The setup of a tensile test involves the installation of a test piece in the load frame of a suitable test machine. Force capacity is the most important factor of a test machine. Other test machine factors, such as calibration and load-frame rigidity, are discussed in more detail in the article "Testing Machines and Strain Sensors" in this Volume. The other aspects of the test setup include proper gripping and alignment of the test piece, and the installation of extensometers or strain sensors when plastic deformation (yield behavior) of the piece is being measured, as described below.

Gripping Devices. The grips must furnish an axial connection between the test piece and the testing machine; that is, the grips must not cause bending in the test piece during loading. The choice of grip is primarily dependent on the geometry of the test piece and, to a lesser degree, on the preference of the test laboratory. That is, rarely do tension-test methods or requirements specify the method of gripping the test pieces.

Figure 24 shows several of the many grips that are in common use, but many other designs are also used. As can be seen, the gripping devices can be classified into several distinct types, wedges, threaded, button, and snubbing. Wedge grips can be used for almost any test-piece geometry; however, the wedge blocks must be designed and installed in the machine to ensure axial loading. Threaded grips and button grips are used only for machined round test pieces. Snubbing grips are used for wire (as shown) or for thin, rectangular test pieces, such as those made from foil.

As shown in Fig. 22, the dimensions of the grip ends for machined round test pieces are usually not specified, and only approximate dimensions are given for the rectangular test pieces. Thus, each test lab must prepare/machine grip ends appropriate for its testing machine. For machined-round test pieces, the grip end is often threaded, but many laboratories prefer either a plain end, which is gripped with the wedges in the same manner as a rectangular test piece, or with a button end that is gripped in a mating female grip. Because the principal disadvantage of a threaded grip is that the pitch of the threads tend to cause a bending moment, a fine-series thread is often used.

Bending stresses are normally not critical with test pieces from ductile materials. However, for test pieces from materials with limited ductility, bending stresses can be important, better alignment may be required. Button grips are often used, but adequate alignment is usually

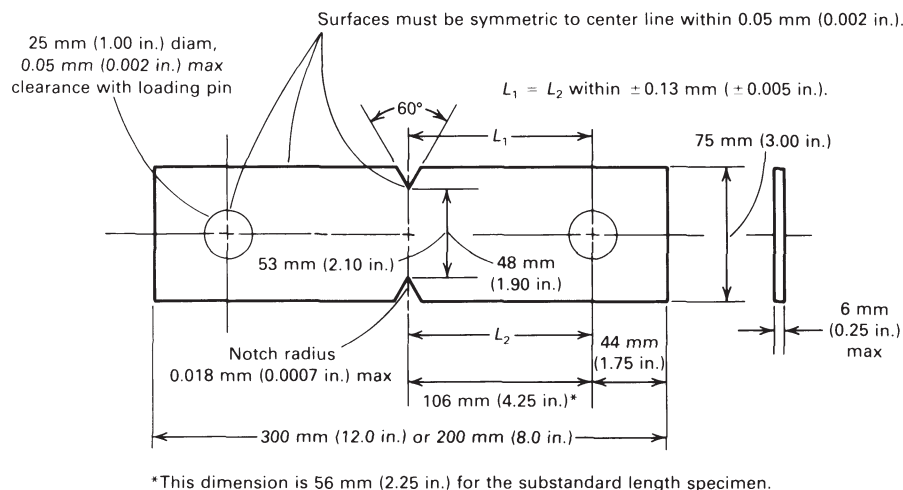


Fig. 23 Example of notched tension-test test piece per ASTM E 338 "Standard Test Method of Sharp-Notch Tension Testing of High-Strength Sheet Materials"

achieved with threaded test pieces. ASTM E 8 also recommends threaded gripping for brittle materials. The principal disadvantage of the button-end grip is that the diameter of the button or the base of the cone is usually at least twice the diameter of the reduced section, which necessitates a larger, rough specimen and more metal removal during machining.

Alignment of the Test Piece. The force-application axis of the gripping device must coincide with the longitudinal axis of symmetry of the test piece. If these axes do not coincide, the test piece will be subjected to a combination of axial loading and bending. The stress acting

on the different locations in the cross section of the test piece then varies, from the sum of the axial and bending stresses on one side of the test piece, to the difference between the two stresses on the other side. Obviously, yielding will begin on the side where the stresses are additive and at a lower apparent stress than would be the case if only the axial stress were present. For this reason, the yield stress may be lowered, and the upper yield stress would appear suppressed in test pieces that normally exhibit an upper yield point. For ductile materials, the effect of bending is minimal, other than the suppression of the upper yield stress. However, if the mate-

rial has little ductility, the increased strain due to bending may cause fracture to occur at a lower stress than if there were no bending.

Similarly, if the test piece is initially bent, for example, coil set in a machined-rectangular cross section or a piece of rod being tested in a full section, bending will occur as the test piece straightens, and the problems exist.

Methods for verification of alignment are described in ASTM E 1012.

Extensometers. When the tension test requires the measurement of strain behavior (i.e., the amount of elastic and/or plastic deformation occurring during loading), extensometers must

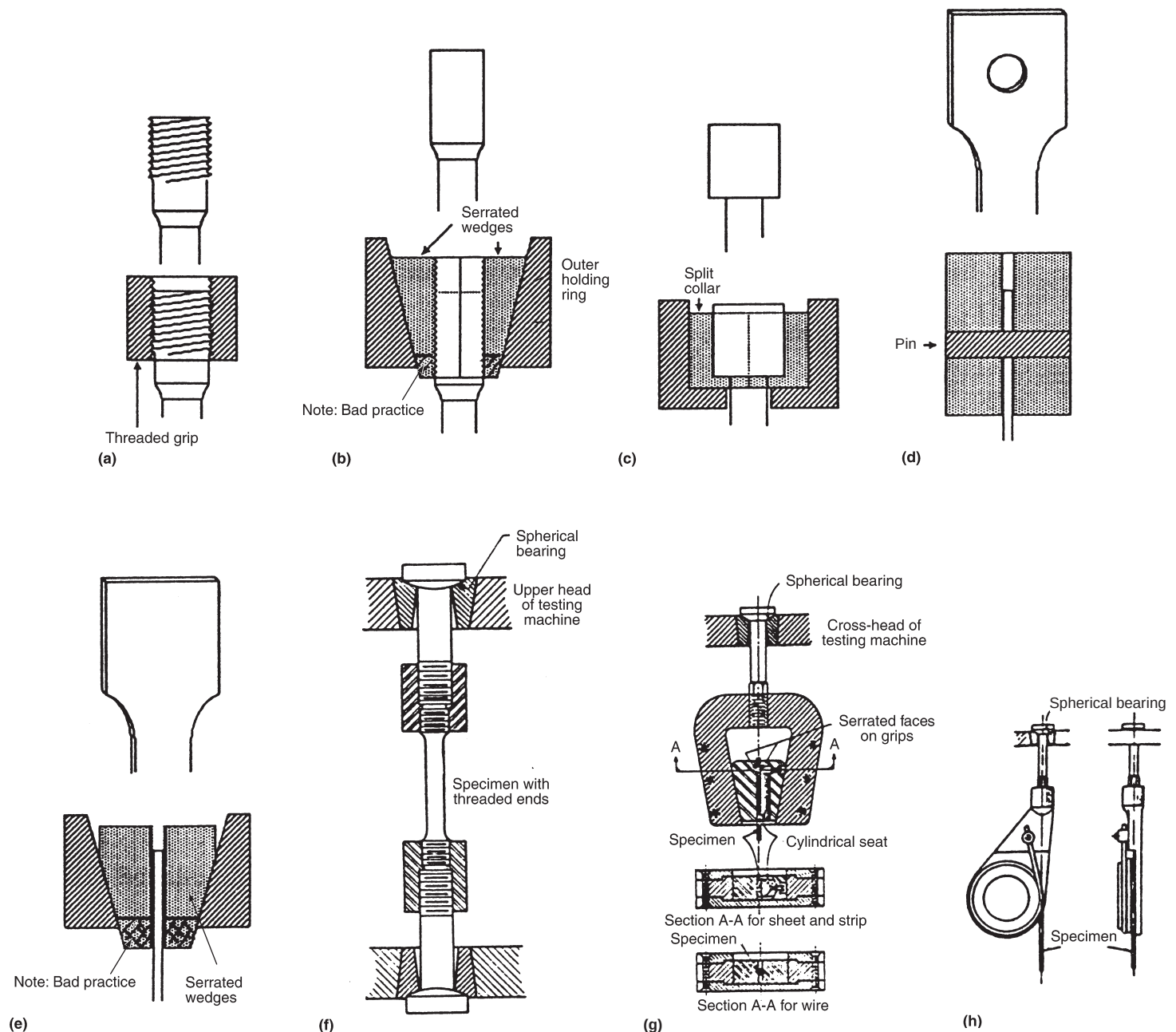


Fig. 24 Examples of gripping methods for tension test pieces. (a) Round specimen with threaded grips. (b) Gripping with serrated wedges with hatched region showing bad practice of wedges extending below the outer holding ring. (c) Butt-end specimen constrained by a split collar. (d) Sheet specimen with pin constraint. (e) Sheet specimen with serrated-wedge grip with hatched region showing the bad practice of wedges extended below the outer holding ring. (f) Gripping device for threaded-end specimen. (g) Gripping device for sheet and wire. (h) Snubbing device for testing wire. Sources: Adapted from Ref 1 and ASTM E 8

be attached to the test piece. The amount of strain can be quite small (e.g., approximately 0.5% or less for elastic strain in steels), and extensometers and other strain-sensing systems are designed to magnify strain measurement into a meaningful signal for data processing.

Several types of extensometers are available, as described in more detail in the article "Testing Machines and Strain Sensors" in this Volume. Extensometers generally have fixed gage lengths. If an extensometer is used only to obtain a portion of the stress-strain curve sufficient to determine the yield properties, the gage length of the extensometer may be shorter than the gage length required for the elongation-at-fracture measurement. It may also be longer, but in general, the extensometer gage length should not exceed approximately 85% of the length of the reduced section or the distance between the grips for test pieces without reduced sections. National and international standardization groups have prepared practices for the classification of extensometers, as described in the article "Testing Machines and Strain Sensors." Extensometer classifications usually are based on error limits of a device, as in ASTM E 83 "Standard Practice for Verification and Classification of Extensometers."

Temperature Control. Tension testing is sometimes performed at temperatures other than room temperature. ASTM E 21 describes standard procedures for elevated-temperature tension testing of metallic materials, which is described further in the article "Hot Tension and Compression Testing" in this Volume. Currently, there is no ASTM standard procedure for cryogenic testing; further information is contained in the article "Tension and Compression Testing at Low Temperatures" in this Volume.

Temperature gradients may occur in temperature-controlled systems, and gradients must be kept within tolerable limits. It is not uncommon to use more than one temperature-sensing device (e.g., thermocouples) when testing at other than room temperature. Besides the tempera-

ture-sensing device used in the control loop, auxiliary sensing devices may be used to determine whether temperature gradients are present along the gage length of the test piece.

Temperature control is also a factor during room-temperature tests because deformation of the test piece causes generation of heat within it. Test results have shown that the heating that occurs during the straining of a test piece can be sufficient to significantly change the properties that are determined because material strength typically decreases with an increase in the test temperature. When performing a test to duplicate the results of others, it is important to know the test speed and whether any special procedures were taken to remove the heat generated by straining the test piece.

Test Procedures

After the test piece has been properly prepared and measured and the test setup established, conducting the test is fairly routine. The test piece is installed properly in the grips, and if required, extensometers or other strain-measuring devices are fastened to the test piece for measurement and recording of extension data. Data acquisition systems also should be checked. In addition, it is sometimes useful to repetitively apply small initial loads and vibrate the load train (a metallographic engraving tool is a suitable vibrator) to overcome friction in various couplings, as shown in Fig. 25(a) and (b). A check can also be run to ensure that the test will run at the proper testing speed and temperature. The test is then begun by initiating force application.

Speed of Testing

The speed of testing is extremely important because mechanical properties are a function of strain rate, as discussed in the section "Effect of

Strain Rate" in this article. It is, therefore, imperative that the speed of testing be specified in either the tension-test method or the product specification.

In general, a slow speed results in lower strength values and larger ductility values than a fast speed; this tendency is more pronounced for lower-strength materials than for higher-strength materials and is the reason that a tension test must be conducted within a narrow test-speed range.

In order to quantify the effect of deformation rate on strength and other properties, a specific definition of testing speed is required. A conventional (quasi-static) tension test, for example, ASTM E 8, prescribes upper and lower limits on the deformation rate, as determined by one of the following methods during the test:

- Strain rate
- Stress rate (when loading is below the proportional limit)
- Cross-head separation rate (or free-running cross-head speed) during the test
- Elapsed time

These methods are listed in order of decreasing precision, except during the occurrence of upper-yield-strength behavior and yield point elongation (YPE) (where the strain rate may not necessarily be the most precise method). For some materials, elapsed time may be adequate, while for other materials, one of the remaining methods with higher precision may be necessary in order to obtain test values within acceptable limits. ASTM E 8 specifies that the

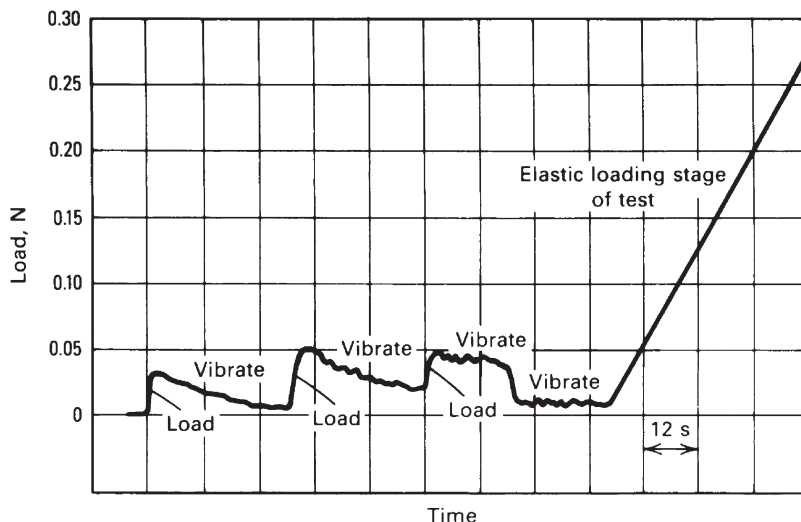


Fig. 25(a) Effectiveness of vibrating the load train to overcome friction in the spherical ball and seat couplings shown in Fig. 25(b)

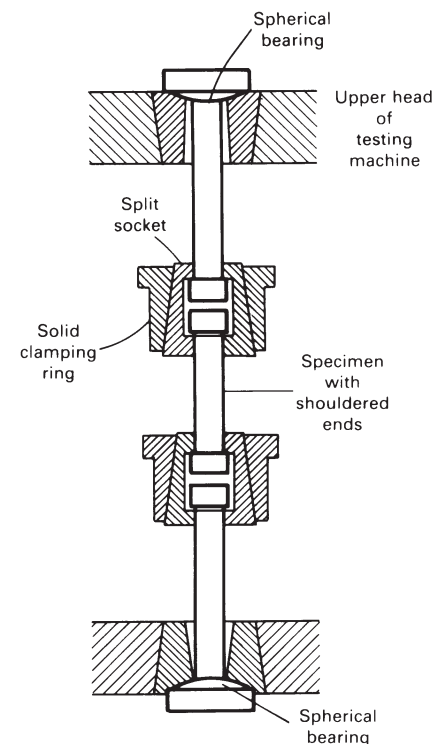


Fig. 25(b) Spherically seated gripping device for shouldered tension test piece

test speed must be slow enough to permit accurate determination of forces and strains. Although the speeds specified by various test methods may differ somewhat, the test speeds for these methods are roughly equivalent in commercial testing.

Strain rate is expressed as the change in strain per unit time, typically expressed in units of min^{-1} or s^{-1} because strain is a dimensionless value expressed as a ratio of change in length per unit length. The strain rate can usually be dialed, or programmed, into the control settings of a computer-controlled system or paced or timed for other systems.

Stress rate is expressed as the change in stress per unit of time. When the stress rate is stipulated, ASTM E 8 requires that it not exceed 100 ksi/min. This number corresponds to an elastic strain rate of about $5 \times 10^{-5} \text{ s}^{-1}$ for steel or $15 \times 10^{-5} \text{ s}^{-1}$ for aluminum. As with strain rate, stress rate usually can be dialed or programmed into the control settings of computer-controlled test systems. However, because most older systems indicate force being applied, and not stress, the operator must convert stress to force and control this quantity.

Many machines are equipped with pacing or indicating devices for the measurement and control of the stress rate, but in the absence of such a device, the average stress rate can be determined with a timing device by observing the time required to apply a known increment of stress. For example, for a test piece with a cross section of 0.500 in. by 0.250 in. and a specified stress rate of 100,000 psi/min, the maximum force application rate would be 12,500 lbf/min (force = stress rate $\times$ area = 100,000 psi/min $\times$ (0.500 in. $\times$ 0.250 in.)). A minimum rate of $\frac{1}{10}$ of the maximum rate is usually specified.

Comparison between Strain-Rate and Stress-Rate Methods. Figure 26 compares strain-rate control with stress-rate control for describing the speed of testing. Below the elastic limit, the two methods are identical. However, as shown in Fig. 26, once the elastic limit is exceeded, the strain rate increases when a constant stress rate is applied. Alternatively, the stress rate decreases when a constant strain rate is specified. For a material with discontinuous yielding and a pronounced upper yield spike (Fig. 7a), it is a physical impossibility for the stress rate to be maintained in that region

because, by definition, there is not a sustained increase in stress in this region. For these reasons, the test methods usually specify that the rate (whether stress rate or strain rate) is set prior to the elastic limit (EL), and the crosshead speed is not adjusted thereafter. Stress rate is not applicable beyond the elastic limit of the material. Test methods that specify rate of straining expect the rate to be controlled during yield; this minimizes effects on the test due to testing machine stiffness.

The rate of separation of the grips (or rate of separation of the cross heads or the cross-head speed) is a commonly used method of specifying the speed of testing. In ASTM A 370, for example, the specification of test speed is that “through the yield, the maximum speed shall not exceed $\frac{1}{16}$ in. per inch of reduced section per minute; beyond yield or when determining tensile strength alone, the maximum speed shall not exceed $\frac{1}{2}$ in. per inch of reduced section per minute. For both cases, the minimum speed shall be greater than $\frac{1}{10}$ of this amount.” This means that for a machined round test piece with a $2\frac{1}{4}$ in. reduced section, the rate prior to yielding can range from a maximum of $\frac{9}{64}$ in./min (i.e., $2\frac{1}{4}$ in. reduced-section length $\times \frac{1}{16}$ in./min) down to $\frac{9}{640}$ in./min (i.e., $2\frac{1}{4}$ in. reduced-section length $\times \frac{1}{160}$ in./min).

The elapsed time to reach some event, such as the onset of yielding or the tensile strength, or the elapsed time to complete the test, is sometimes specified. In this case, multiple test pieces are usually required so that the correct test speed can be determined by trial and error.

Many test methods permit any speed of testing below some percentage of the specified yield or tensile strength to allow time to adjust the force application mechanism, ensure that the extensometer is working, and so on. Values of 50 and 25%, respectively, are often used.

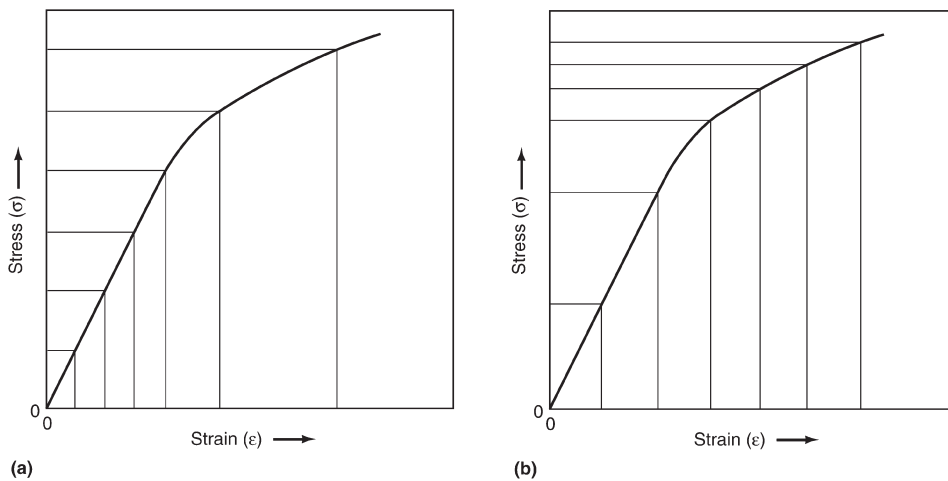


Fig. 26 Illustration of the differences between constant stress increments and constant strain increments. (a) Equal stress increments (increasing strain increments). (b) Equal strain increments (decreasing stress increments)

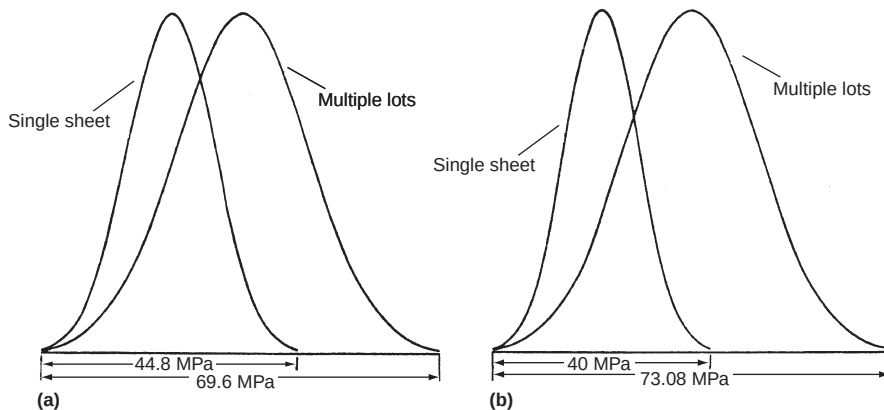


Fig. 27 Distribution of (a) yield and (b) tensile strengths for multiple tests on single sheet and on multiple lots of aluminum alloy 7075-T6. Source: Ref 10

Post-Test Measurements

After the test has been completed, it is often required that the cross-sectional dimensions again be measured to obtain measures of ductility. ASTM E 8 states that measurements made after the test shall be to the same accuracy as the initial measurements.

Method E 8 also states that upon completion of the test, gage lengths 2 in. and under are to be measured to the nearest 0.01 in., and gage lengths over 2 in. are to be measured to the nearest 0.5%. The document goes on to state that a percentage scale reading to 0.5 % of the gage length may be used. However, if the tension test is being performed as part of a product specification, and the elongation is specified to meet a value of 3% or less, special techniques, which are described, are to be used to measure the final gage length. These measurements are discussed in a previous section, “Elongation,” in this article.

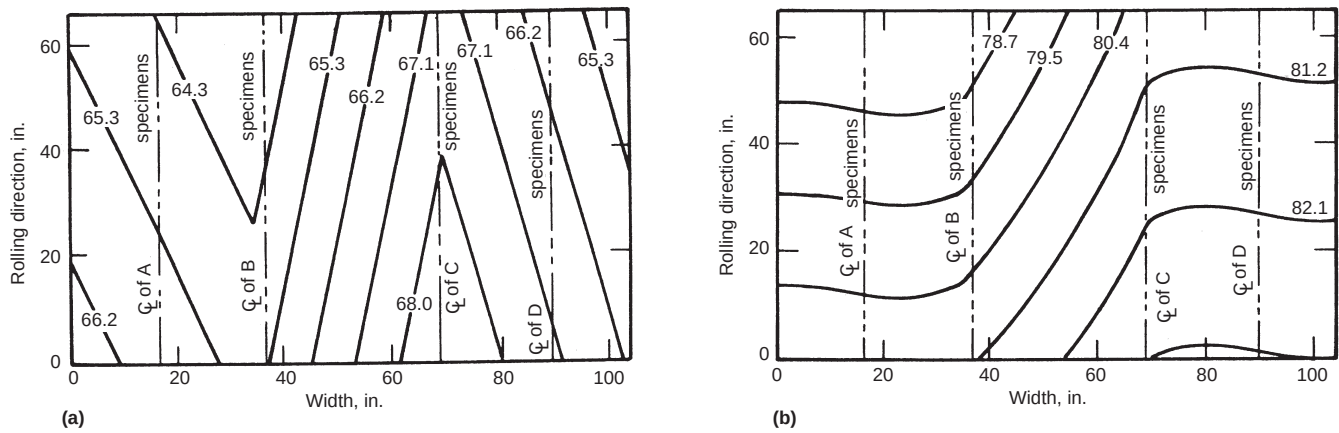


Fig. 28 Contour maps of (a) constant yield strength (0.5% elongation under load, ksi) and (b) constant tensile strength (ksi) for a plate of alloy steel

Variability of Tensile Properties

Even carefully performed tests will exhibit variability because of the nonhomogenous nature of metallic materials. Figure 27 (Ref 10) shows the three-sigma distribution of the offset yield strength and tensile strength values that were obtained from multiple tests on a single aluminum alloy. Distribution curves are presented for the results from multiple tests of a single sheet and for the results from tests on a

number of sheets from a number of lots of the same alloy. Because these data are plotted with the minus three-sigma value as zero, it appears there is a difference between the mean values; however, this appearance is due only to the way the data are presented. Figures 28(a) and (b) show lines of constant offset yield strength and constant tensile strength, respectively, for a 1 in. thick, quenched and tempered plate of an alloy steel. In this case, rectangular test pieces $1\frac{1}{2}$ in. wide were taken along the transverse direction (T orientation in Fig. 20) every 3 in. along each of the four test-piece centerlines shown. These data indicate that the yield and tensile strengths vary greatly within this relatively small sample and that the shape and location of the yield strength contour lines are not the same as the shape and location of the tensile strength lines.

Effect of Strain Concentrations. During testing, strain concentrations (often called stress concentrations) occur in the test piece where there is a change in the geometry. In particular, the transition radii between the reduced section and the grip ends are important, as previously noted in the section on test-piece geometry. Most test methods specify a minimum value for these radii. However, because there is a change in geometry, there is still a strain concentration at the point of tangency between the

radii and the reduced section. Figure 29(a) (Ref 11) shows a test piece of rubber with an abrupt change of section, which is a model of a tension test piece in the transition region. Prior to applying the force at the ends of the model, a rectangular grid was placed on the test piece. When force is applied, it can be seen that the grid is severely distorted at the point of tangency but to a much lesser degree at the center of the model. The distortion is a visual measure of strain. The strain distribution across section $n-n$ is plotted in Fig. 29(b). From the stress-strain curve for the material (Fig. 29c), the stresses on this section can be determined. It is apparent that the test piece will yield at the point of tangency prior to general yielding in the reduced section. The ratio between the nominal strain and actual, maximum strain is often referred to as the strain-concentration factor, or the stress-concentration factor if the actual stress is less than the elastic limit. This ratio is often abbreviated as k_t . Studies have shown that k_t is about 1.25 when the radii are $\frac{1}{2}$ in., the width (or diameter) of the reduced section is 0.500 in., and the width (or diameter) of the grip end is $\frac{3}{4}$ in. That is, the actual strain or the actual elastic stress at the transition (if less than the yield of the material) is 25% greater than would be expected without consideration of the strain or

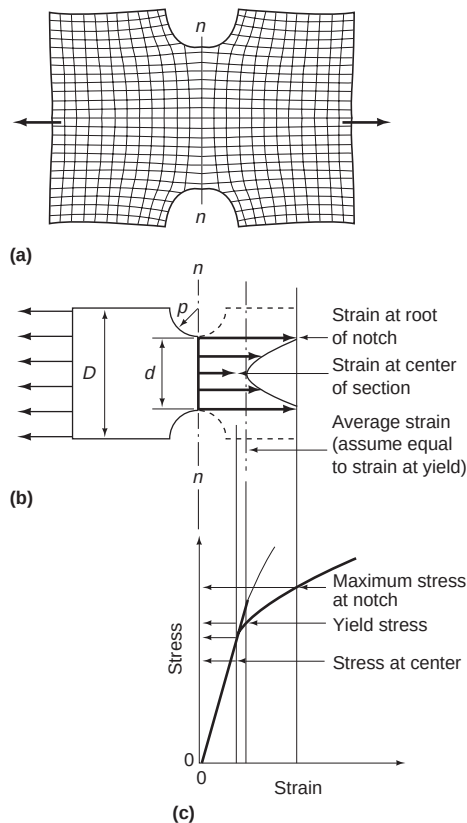


Fig. 29 Effect of strain concentrations on section $n-n$. (a) Strain distribution caused by an abrupt change in cross section (grid on sheet of rubber) (Ref 11). (b) Schematic of strain distribution on cross section $n-n$ (Ref 11). (c) Calculation of stresses at abrupt change in cross section $n-n$ by graphical means

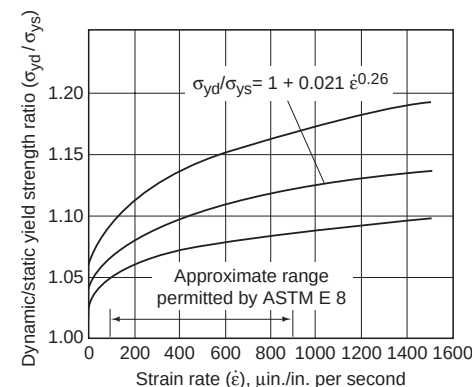


Fig. 30 Effect of strain rate on the ratio of dynamic yield-stress and static yield-stress level of A36 structural steel. Source: Ref 12

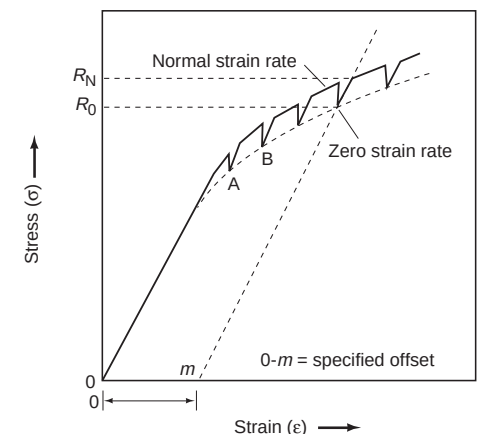


Fig. 31 Stress-strain curves for tests conducted at "normal" and "zero" strain rates

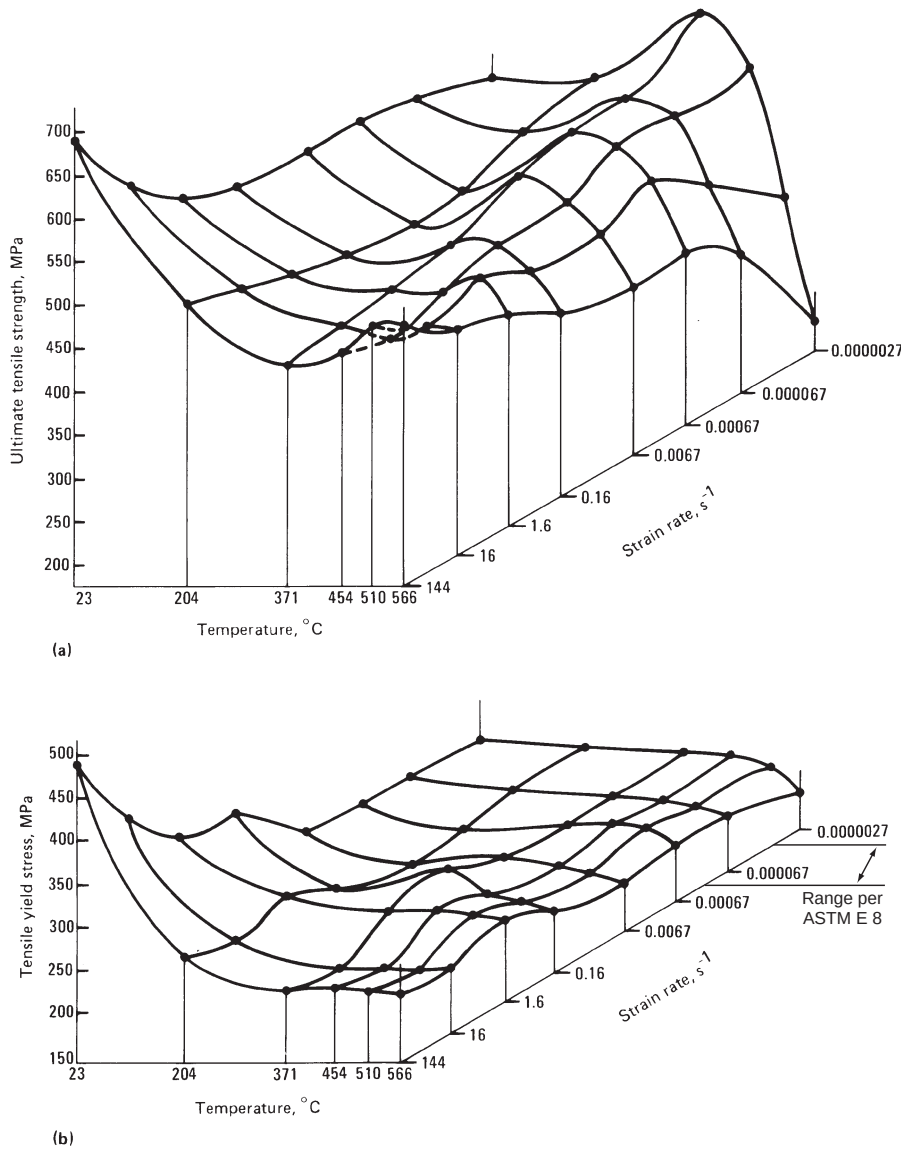


Fig. 32 Effect of temperature and strain rate on (a) tensile strength and (b) yield strength of 2 1/4 Cr-1 Mo Steel. Note: Strain-rate range permitted by ASTM Method E8 when determining yield strength at room temperature is indicated. Source: Ref 13

stress concentration. The value of k_t decreases as the radii increase such that, for the above example, if the radii are 1.0 in., and k_t decreases to about 1.15.

Various techniques have been tried to minimize k_t , including the use of spirals instead of radii, but there will always be strain concentration in the transition region. This indicates that the yielding of the test piece will always initiate at this point of tangency and proceed toward midlength. For these reasons, it is extremely important that the radii be as large as feasible when testing materials with low ductility.

Strain concentrations can be caused by notches deliberately machined in the test piece, nicks from accidental causes, or shear burrs, machining marks, or gouges that occur during the preparation of the test piece or from many other causes.

Effect of Strain Rate. Although the mechanical response of different materials varies, the strength properties of most materials tend to increase at higher strain rates. For example, the variability in yield strength of ASTM A 36 structural steel over a limited range of strain rates is shown in Fig. 30 (Ref 12). A “zero-strain-rate” stress-strain curve (Fig. 31) is generated by applying forces to a test piece to obtain a small plastic strain and then maintaining that strain until the force ceases to decrease (Point A). Force is reapplied to the test piece to obtain another increment of plastic strain, which is maintained until the force ceases to decrease (Point B). This procedure is continued for several more cycles. The smooth curve fitted through Points A, B, and so on is the “zero-strain-rate” stress-strain curve, and the yield value is determined from this curve.

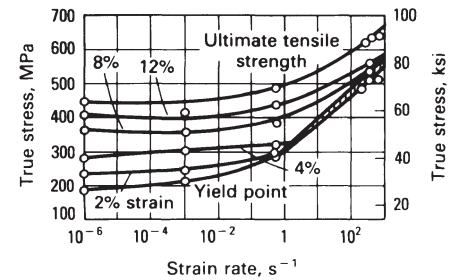


Fig. 33 True stresses at various strains vs. strain rate for a low-carbon steel at room temperature. The top line in the graph is tensile strength, and the other lines are yield points for the indicated level of strain. Source: Ref 14

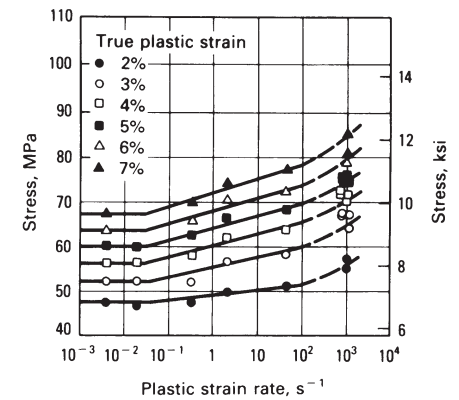


Fig. 34 Uniaxial stress/strain/strain rate data for aluminum 1060-O. Source: Ref 16

The effect of strain rate on strength depends on the material and the test temperature. Figure 32 (Ref 13) shows graphs of tensile strength and yield strength for a common heat-resistant low-alloy steel (2 1/4 Cr-1 Mo) over a wide range of temperatures and strain rates. In this figure, the strain rates were generally faster than those prescribed in ASTM E 8.

Another example of strain effects on strength is shown in Fig. 33 (Ref 14). This figure illustrates true yield stress at various strains for a low-carbon steel at room temperature. Between strain rates of 10⁻⁶ s⁻¹ and 10⁻³ s⁻¹ (a thousandfold increase), yield stress increases only by 10%. Above 1 s⁻¹, however, an equivalent rate increase doubles the yield stress. For the data in Fig. 33, at every level of strain the yield stress increases with increasing strain rate. However, a decrease in strain-hardening rate is exhibited at the higher deformation rates. For a low-carbon steel tested at elevated temperatures, the effects of strain rate on strength can become more complicated by various metallurgical factors such as dynamic strain aging in the “blue brittleness” region of some mild steels (Ref 14).

Structural aluminum is less strain-rate sensitive than steels. Figure 34 (Ref 15) shows data obtained for 1060-O aluminum. Between strain rates of 10⁻³ s⁻¹ and 10³ s⁻¹ (a millionfold increase), the stress at 2% plastic strain increases by less than 20%.

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